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Text Encoding in Quantum Natural Language Processing: A Systematic Survey

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ABSTRACT Quantum natural language processing (QNLP) applies superposition, entanglement, and interference to the representation and processing of linguistic information. Any QNLP pipeline, whatever methods it uses, must first map discrete or distributed text representations onto qubit registers or parameterized-circuit inputs. This encoding step governs model expressivity, qubit resource requirements, and trainability. Yet it is rarely studied as an independent variable, and no standard method brings these evaluation aspects together for analyzing and comparing encodings. This article aims to fill this gap by introducing a classification scheme for text encodings, covering both the encoding used and the experimental evidence reported for it, and employs it to carry out a systematic literature review of text encoding strategies in QNLP. The scope is limited to the encoding step: circuit design, ansatz expressibility, and error correction are covered only where they interact directly with encoding choices. The survey covers 125 studies published between 2021 and 2026, using a protocol based on state-of-the-art guidelines to define the search strategy, eligibility criteria, screening procedure, snowball sampling, and data extraction. The scheme is organized around a unifying taxonomy for analyzing and comparing encodings. Seven encoding families are identified and characterized: four major families, angle, amplitude, syntactic (DisCoCat), and structured feature maps (IQP/ZZ), and three emerging families: data re-uploading, hybrid classical–quantum embedding, and Hamiltonian/observable encoding. Their distribution is analyzed across NLP tasks, simulation and hardware environments, qubit budgets, and reporting practice. Three findings stand out: noiseless circuit simulation dominates the literature, real-hardware execution remains rare, and evaluation is too fragmented across author-chosen datasets, baselines, and metrics for results to be compared between studies. The encoding bottleneck, the dimensionality mismatch between linguistic representations and near-term qubit capacities, is stated explicitly as a basis for resource-efficient QNLP algorithm and hardware design.

INDEX TERMS Encoding bottleneck, parameterized quantum circuits, quantum machine learning, quantum natural language processing, systematic literature review, text encoding.

I. INTRODUCTION

EVERY quantum natural language processing (QNLP) system begins with a mandatory step: converting text into a quantum state. Before a single gate is applied or a single parameter is trained, a token, sentence, or document must be mapped onto qubits. This mapping, the encoding, determines how many qubits are consumed and how deep the resulting circuit must be. It determines whether gradients can be computed efficiently, and whether any genuinely quantum property, such as entanglement, is brought to bear on the linguistic content. It is the most consequential design decision in the pipeline.

Despite this centrality, encoding is rarely the subject of dedicated study: effort concentrates instead on other stages of the pipeline, such as ansatz design [1], [2] and the choice of classical optimizer [3], or on carrying QNLP to a new task or domain [4], [5]. Most QNLP papers dispatch it in a paragraph, introduce one strategy without justification, and move on to reporting accuracy numbers. After five years of accelerating publication activity, the result is a fragmented body of work: dozens of encoding families coexist, yet there is no common approach for analyzing and comparing them.

Unified taxonomies, systematic comparisons, and a shared understanding of encoding trade-offs are all missing from the literature. Existing QNLP surveys and reviews do not isolate encoding as a variable [6], [7]. The few explicitly comparative studies contrast model families or ansätze rather than encoding schemes [8], [9]. This article fills that gap: it conducts an extensive systematic literature review of the topic and makes text encoding in QNLP the primary subject of study.

A. SCOPE

This survey reviews research that (1) applies a genuine quantum circuit (simulated noiseless, simulated noisy, or executed on real hardware) to at least one NLP task, and (2) identifies, at least by name or type, the encoding scheme used to map text or text-derived features onto qubits or circuit parameters. The review follows an SLR protocol defined before the search (Section III) adapted from established guidelines for systematic literature reviews in computer science [10].

Papers that use quantum-inspired or quantum-motivated classical algorithms without instantiating an actual parameterized quantum circuit are excluded. Also excluded are survey and review papers, hardware-focused benchmarking studies that treat NLP datasets as a generic payload, and papers whose methodological detail is insufficient to identify the encoding scheme, according to the inclusion and exclusion criteria stated in Section III-B. The final corpus comprises 125 studies.

The scope is limited to the encoding step. Quantum circuit design, ansatz expressibility, and quantum error correction are surveyed only where they interact directly with encoding choices. Classical NLP methods are discussed only as baselines.

B. RESEARCH QUESTIONS

The review is organized around five research questions, each answered from the data extracted across the corpus rather than from the claims of individual studies:

RQ1) Which text encoding strategies are used in QNLP, and by what principle can they be organized into families? (Section IV)
RQ2) How does the use of each family vary across model families and ansätze, and can an encoding be compared independently of the architecture hosting it? (Section V)
RQ3) How has the mix of encodings and architectures changed over the years the corpus spans? (Section VI)
RQ4) Does the choice of encoding track the NLP task, and is there a task family whose structure gives a principled reason to prefer one encoding over another? (Section VII)
RQ5) How rigorously are encoding claims evaluated, in terms of execution regime, hardware validation, and reproducibility? (Section VIII)

C. KEY CONTRIBUTIONS

This article makes four contributions to the study of text encoding in QNLP:

1) A novel taxonomy of text encoding strategies, organized by the point at which text enters the quantum computation and covering seven encoding families (four major and three emerging) with their specialized variants.
2) A systematic review of 125 studies that apply or propose quantum text encodings across NLP tasks, conducted under a protocol defined before the search and grounded in established methodological guidelines for systematic literature reviews: the Keele and Charters procedure for systematic reviews in computing [10], the PRISMA 2020 statement for reporting the selection process [11], and Wohlin's guidelines for snowball sampling [12].
3) A quantitative cross-study characterization of the corpus using the classification scheme defined in Section III-F, spanning encoding-by-model-family, temporal, task-fit, and experimental-rigor analyses.
4) A characterization of the encoding bottleneck in light of the survey results: the gap between the dimensionality of the text representations studies start from and the qubit budgets they actually use is quantified over the corpus, and each encoding family is read as a choice of which cost absorbs that gap, discarded information, added circuit depth, or a classical burden relocated upstream. This characterization, together with the gaps documented across the review, yields a corpus-derived set of five open challenges and an actionable agenda for future work.

D. RELATED WORK

Several prior works survey the QNLP literature, but none treats text encoding as the central variable. Varmantchaonala et al. [13] provide the most comprehensive QNLP survey to date, covering quantum algorithms, circuit architectures, and NLP task categories; encoding appears only as an implementation detail. Guarasci et al. [14] offer an early and influential overview of QNLP challenges and opportunities, from theoretical quantum-advantage arguments to hardware experiments, without isolating encoding as a dimension of comparison. Widdows et al. [15] examine word embeddings and sequential quantum models with more attention to representation trade-offs than any other general survey, but stop short of a systematic comparison. Bouakba and Belhadef [8] experimentally compare compositional DisCoCat models against quantum neural networks; their compositional-versus-flat-vector axis overlaps with the taxonomy proposed here, though their scope is limited to two model families. Huzzat et al. [16] is the most encoding-aware prior work: it addresses the mismatch between text-representation dimensionality and the qubit capacity of NISQ classifiers, and treats classical compression as a prerequisite for stable training. It

remains, however, a conceptual primer for a single model class rather than a systematic review.

Beyond QNLP itself, the quantum engineering literature has an established tradition of comprehensive subfield surveys, covering, for instance, quantum circuit compilation for NISQ devices [17], quantum computing for finance [18], and entanglement routing in quantum networks [19], the last of which likewise organizes its field through an explicit taxonomy of approaches. None of these addresses text encoding in QNLP, which is the gap the present survey fills within that tradition.

This survey complements these works by treating encoding as the primary unit of analysis, yielding the first quantitative cross-study characterization of how text is encoded in QNLP and at what resource cost.

E. STRUCTURE

The rest of the article is organized as follows. Section II provides the background on quantum computing, compositional distributional semantics, and the role of encoding in QNLP. Section III describes the systematic review protocol (search strategy, eligibility criteria, screening, and snowball sampling) and the data extraction and classification scheme used throughout the analysis. Sections IV–VIII then answer RQ1–RQ5 in turn. Section IX synthesizes the trends, states the encoding bottleneck explicitly, and enumerates the open challenges and the future-work agenda. Section X concludes.

II. BACKGROUND

A. QUANTUM COMPUTING PRIMITIVES

A quantum bit (qubit) occupies a superposition of the basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \tag{1}$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes. A register of n qubits occupies a 2^n-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^{2^n}$; its state is a unit vector in that space. This exponential dimension is the property every amplitude-style encoding exploits. Entanglement introduces correlations with no classical counterpart, and measurement collapses the state to a classical bitstring, destroying superposition. Whether superposition, entanglement, or interference confers a practical benefit on NLP tasks remains the central open question of the field, and Section VIII examines how far the surveyed evidence can speak to it.

B. PARAMETERIZED QUANTUM CIRCUITS

The dominant computational model in near-term quantum machine learning is the parameterized quantum circuit (PQC), also called a variational quantum circuit [20]. A PQC applies a sequence of unitary gates to an initial state $|\psi_0\rangle$:

$$U(\boldsymbol{\theta}) = U_L(\boldsymbol{\theta}_L) \cdots U_1(\boldsymbol{\theta}_1), \tag{2}$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ are trainable parameters. Model outputs are expectation values $\langle O \rangle = \langle \psi_0 | U^\dagger(\boldsymbol{\theta})\, O\, U(\boldsymbol{\theta}) | \psi_0 \rangle$, and gradients follow analytically from the parameter-shift rule [21], [22]. PQCs underlie six of the seven encoding families surveyed here; the exception is grammatical encoding, which derives circuit structure from sentence syntax rather than from a fixed ansatz.

PQCs operate under the constraints of noisy intermediate-scale quantum (NISQ) hardware [23]: shallow circuits, limited qubit counts, and significant gate error rates. These constraints bear directly on the encoding step. Every qubit or gate layer spent loading a text representation is a resource unavailable to the learning task.

C. TEXT ENCODING

Text encoding is the procedure that maps a text input x (a token, sentence, or document) to a quantum state $|\phi(x)\rangle \in \mathcal{H}$ or to a parameter vector $\boldsymbol{\theta}(x)$ that initializes a PQC. It is the mandatory interface between classical language data and the quantum circuit. The input x may be a raw token sequence, a classical embedding vector from a pre-trained model, or a parse tree carrying grammatical structure. Which representation is used, and how it is loaded onto qubits, defines the encoding family.

D. COMPOSITIONAL DISTRIBUTIONAL SEMANTICS AND DISCOCAT

One of the two main traditions in QNLP originates in categorical linguistics. Coecke, Sadrzadeh, and Clark [24] showed that the compact closed categorical structure of pregroup grammars also governs quantum processes in Hilbert space. In a pregroup grammar, each word carries a grammatical type, and a sentence is parsed by reducing compound types through tensor contractions. These reductions obey the same diagrammatic calculus as quantum processes, so a parse tree translates directly into a quantum circuit: words become parameterized quantum states, and grammatical reductions become cups in a string diagram.

This is the DisCoCat (Distributional Compositional Categorical) model, operationalized as an end-to-end pipeline by [25] and executed on near-term hardware in [26], [27] via the lambeq toolkit. The circuit structure is not fixed in advance: the parse tree determines it, so syntactically distinct sentences produce structurally distinct circuits. This property sets the grammatical encoding family apart from all other families, which treat the sentence as a flat feature vector.

E. QUANTUM MACHINE LEARNING FOR TEXT

The second tradition approaches NLP as a classification problem. It applies quantum feature maps [28] and variational circuits [20] to text-derived feature vectors: classical embeddings (word2vec, GloVe, or transformer outputs) are compressed to the available qubit count, loaded into a PQC via rotation gates or amplitude encoding, and trained end-to-end on labeled examples. Because this approach assumes nothing about grammatical structure, it applies to arbitrary text without a parser and scales more easily to realistic

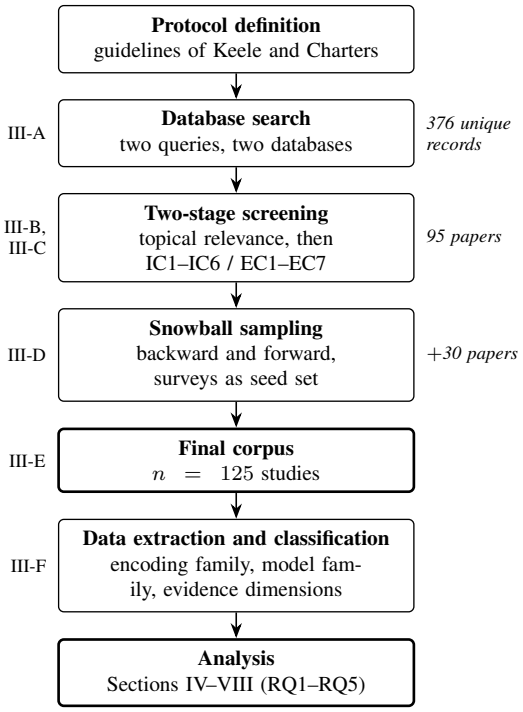


FIGURE 1. Overview of the review workflow. Labels on the left give the subsection in which each step is described; Fig. 2 gives the record-level detail of the search, screening, and snowball stages.

dataset sizes. Its logical endpoint is the hybrid quantum–classical architecture, in which a large classical language model handles high-dimensional representation and a small quantum circuit acts as a downstream head; such architectures have grown in prevalence since 2023.

The two traditions differ in how they encode text and in what they claim the quantum component contributes. Compositional models argue for structural alignment between grammar and quantum computation; QML models argue for expressivity or kernel separability.

III. REVIEW METHODOLOGY

This systematic literature review followed the guidelines for systematic literature reviews in software engineering established by Keele and Charters [10]. The protocol was defined before the search was executed. It fixed the search strategy, eligibility criteria, screening procedure, snowball sampling, and data extraction process. Fig. 1 gives an overview of the resulting workflow and of where each step is described below.

A. SEARCH STRATEGY

Two complementary search queries were executed on two electronic databases, Scopus and IEEE Xplore, to cover both the quantum machine learning and the compositional QNLP literature. The first query targeted the intersection of quantum computing or machine learning with NLP and encoding-related terms. The second targeted QNLP-specific terminology: the DisCoCat and DisCoCirc frameworks, established pipelines that implement them such as the lambeq toolkit, and categorical linguistic concepts. The searches were restricted to English-language journal articles and conference papers published between 2021 and 2026 (ends included), excluding editorials, book chapters, and erratum notices. They returned 269 results from Scopus and 217 from IEEE Xplore, a combined total of 486 records. Removing 110 duplicates left 376 unique records.

B. ELIGIBILITY CRITERIA

A study was selected if and only if it met all six inclusion criteria below.

- *IC1: NLP task grounding.* The study applies or evaluates the proposed method on at least one natural language processing task, such as classification, similarity, sequence labeling, question answering, or generation.
- *IC2: Genuine quantum computation.* The study implements and evaluates the method in at least a noiseless quantum circuit simulation (exact statevector simulation or equivalent) or on real quantum hardware; merely quantum-inspired classical algorithms, such as tensor-network models or classical approximations of quantum kernels, do not satisfy this criterion.
- *IC3: Encoding scheme identified.* The study names or identifies at least the type of encoding used to map text or text-derived features onto qubits, quantum states, or circuit parameters; a named or typed mention is sufficient, and circuit diagrams or full mathematical derivations are not required.
- *IC4: Peer-reviewed or preprint venue.* The study appeared in a peer-reviewed journal, conference, or workshop, or as a preprint with sufficient methodological detail to assess IC1–IC3.
- *IC5: Language and access.* The study is available in full text, in English, and accessible for review.
- *IC6: Time window.* The study was published between 2021 and 2026 inclusive.

Even if a study satisfied all six inclusion criteria, it was excluded if any of the following applied:

- *EC1: No NLP application.* The study applies quantum machine learning exclusively to non-text domains and offers no transferable text-encoding discussion.
- *EC2: Classical-only encoding.* The study uses encoding purely in the classical sense (tokenization, positional encoding, one-hot) with no quantum circuit involved.
- *EC3: Insufficient detail.* The study is an abstract, extended abstract, or short paper that names an encoding but provides no formal description or reproducible detail.
- *EC4: Duplicate or superseded.* The study is a conference version superseded by an extended journal version; only the more complete version was retained.
- *EC5: Pure hardware focus.* The study addresses hardware benchmarking alone and uses an NLP dataset only

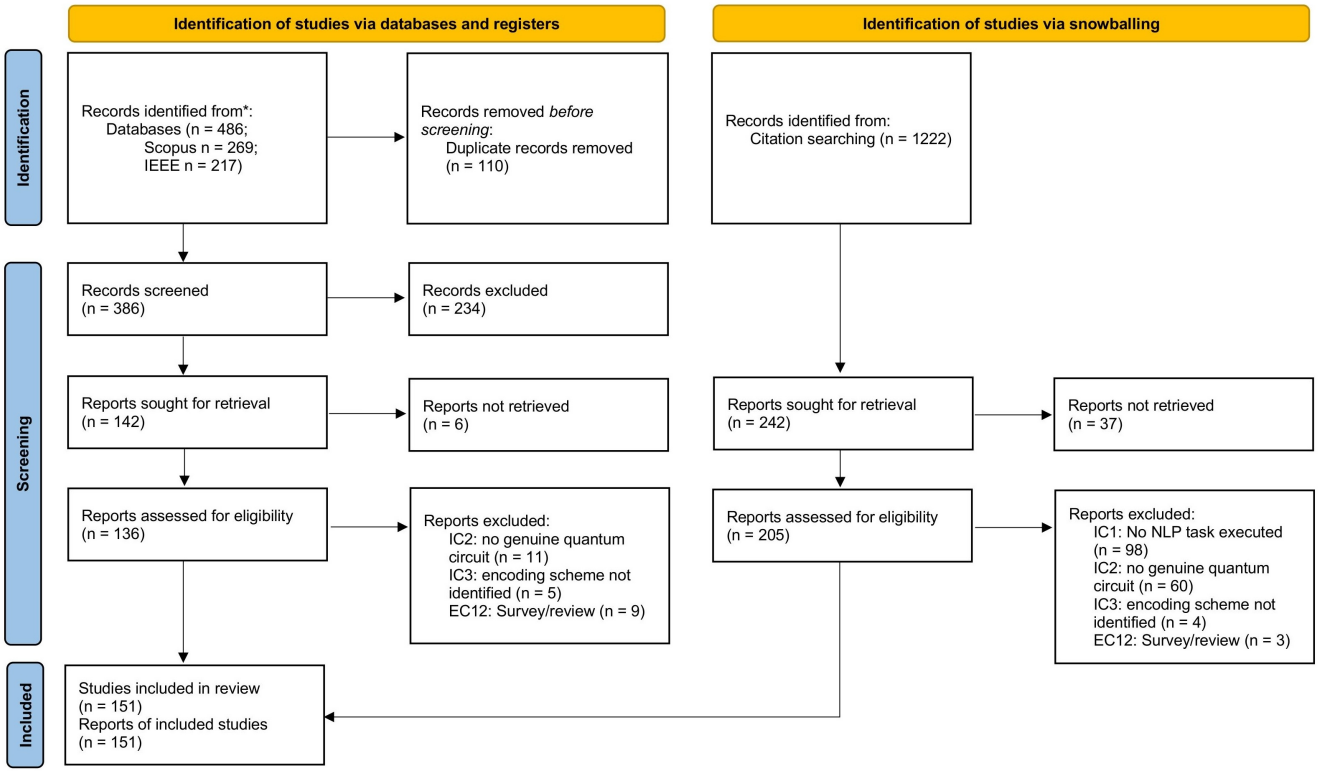


FIGURE 2. PRISMA flow diagram. Final corpus: $n = 125$ (95 from database search, 30 from snowball sampling).

as a generic payload, with no discussion of linguistic encoding properties.

- *EC6: Quantum-inspired, classical surrogate, or unverifiable simulation.* The study uses quantum-derived mathematical formalism without instantiating an actual parameterized quantum circuit, reports only classical surrogates of quantum behavior, or claims quantum simulation without evidence of framework or circuit use.
- *EC7: Survey or review paper.* The study is primarily a survey or review of existing work rather than an original research contribution; such papers were retained as reference material and are discussed in Section I-D.

C. SCREENING AND SELECTION

Screening proceeded in two stages. In the first stage, all 376 records were assessed on title and abstract alone for broad topical relevance to quantum computing applied to text. No IC/EC criteria were applied at this stage; its only purpose was to remove records clearly outside the scope of the review. This eliminated 234 records and left 142 reports sought for retrieval, of which 6 could not be retrieved, leaving 136 reports assessed for eligibility.

In the second stage, the full text of each of the 136 reports was assessed against the complete set of inclusion and exclusion criteria of Section III-B. This yielded 95 papers for the corpus. The remaining 41 retrieved reports were excluded or retained as surveys, mainly under IC2 (no genuine quantum circuit; $n = 15$), IC3 (encoding scheme not identified; $n = 6$), EC6 (quantum-inspired, classical surrogate, or unverifiable simulation; $n = 3$), and EC7 (survey or review; $n = 11$); six records were excluded under other criteria.

D. SNOWBALL SAMPLING

To reduce the risk of missing relevant work not indexed by the two databases, two rounds of backward and forward snowball sampling [12] were applied to the surveys excluded under EC7. Surveys were chosen as the seed set, rather than the included studies, for two reasons. First, Wohlin's guidelines ask for a start set that is small, diverse, and well connected to the target population; in a field as young as QNLP, the reference list of a survey is by construction a curated, near-exhaustive sample of the literature in the field up to its search date, so each survey seed contributes far more unique candidates than an individual study, whose citations concentrate on the same few foundational works. Second, forward snowballing from surveys exploits a stable citation practice: new studies routinely cite the reviews of the field, so the papers citing a survey are enriched for precisely the recent work that database indexing lags behind. The cost of this choice is that a study neither cited by nor citing any survey of the field remains invisible to the procedure; this residual risk is discussed in Section X. Reference lists and citing papers yielded 1222 unique candidate records. Title-and-abstract screening for broad topical relevance reduced

this set to 242 reports sought for retrieval, of which 37 could not be retrieved, leaving 205 reports assessed for eligibility. Full-text assessment against the complete IC/EC protocol then added 30 papers to the corpus. The remaining 175 retrieved reports were excluded or retained as surveys. The main grounds were IC1 (no NLP task; $n = 74$), IC2 (no genuine quantum circuit; $n = 52$), IC6 (outside the time window; $n = 20$), IC3 (encoding scheme not identified; $n = 4$), and EC7 (survey or review; $n = 22$); three records were excluded under other criteria.

E. FINAL CORPUS

The final corpus comprises 125 papers, 95 from the database search and 30 from snowball sampling; Fig. 2 summarizes the selection process as a PRISMA 2020 flow diagram [11].

F. DATA EXTRACTION AND CLASSIFICATION

Each study in the corpus was classified along a fixed set of dimensions, so that every classification, cross-tabulation, and trend reported in Sections IV–IX follows directly from the extraction sheet rather than from an ad hoc reading of the corpus. The two primary dimensions are the encoding family, defined by the point at which text or text-derived features enter the quantum computation (Section IV), and the model family, defined by the architecture that consumes the encoded state. Secondary dimensions capture the resources and evidence attached to that choice: qubit count and whether qubit scaling was tested, execution regime, presence of a classical baseline, and whether scalability was discussed. Each dimension is operationalized as one or more columns of the structured extraction sheet compiled during data extraction.

Assignment follows explicit decision rules. A study is classified by its dominant encoding interface: the one that carries the text representation into the circuit whose output is evaluated. Auxiliary encodings used only in ablations or baselines do not change the class; a study whose main pipeline is amplitude-encoded but that reports an angle-encoded ablation is classified as amplitude. Where a pipeline stacks a classical encoder in front of a quantum layer, the study is classified as `hybrid_embed` only when the classical stage is co-trained or purpose-built as part of the contribution; a frozen pretrained embedding feeding a standard rotation layer remains `angle`. The `hybrid_nn` model family covers any architecture in which a classical deep network (transformer, recurrent, or convolutional, not limited to LLM-based extractors) performs the main representational work and the quantum circuit is a submodule. A bare variational classifier over precomputed features is `vqc`, even when those features come from a deep model. Compositional (DisCoCat-style) studies are identified by circuit topology derived from a grammatical parse, regardless of toolkit. These disambiguation rules were fixed during protocol definition and applied uniformly during extraction.

Extraction and classification were performed by one author and reviewed by a second; disagreements were resolved by discussion against the decision rules above. All figures below are reported over the full corpus of 125 studies.

The corpus is small enough that many of the cross-tabulations below have cells in the low single digits, and every count is a complete enumeration of the corpus rather than an estimate from a sample of it. All distributions are therefore reported descriptively, as counts and shares of a known population. Where the article asserts that two dimensions are coupled rather than simply describing one, the claim rests on the size of the imbalance in the cross-tabulation together with a stated mechanism, and both are given. Percentages computed on denominators below ten are reported with their denominators and are not read as evidence of a difference.

IV. TAXONOMY OF TEXT ENCODING STRATEGIES

This section answers RQ1. The encoding problem in QNLP is not a single operation. Depending on the study, text can enter the quantum computation as rotation parameters, as amplitudes of a prepared state, as a grammar-derived circuit, as a structured feature map, or through a more specialized interface such as repeated data injection, a hybrid neural representation, or an observable. These mechanisms are often discussed under the same term, "encoding," although they make different assumptions about where linguistic information is represented and which quantum resource is being used.

Table 1 organizes these mechanisms into the proposed taxonomy, applying the classification rules of Section III-F: every study discussed below is assigned to exactly one encoding family by the dominant-interface rule, based on the point at which text or text-derived features enter the quantum computation. The optimizer, measurement rule, downstream classifier, and ansatz are treated as design choices that can be combined with any encoding family.

The taxonomy comprises seven encoding families. Four are major: angle encoding ($n = 51$), amplitude encoding ($n = 28$), syntactic/DisCoCat encoding ($n = 29$), and structured feature-map encoding ($n = 9$). Three are emerging families that together account for the remaining eight studies: data re-uploading ($n = 4$), hybrid classical–quantum embedding ($n = 3$), and Hamiltonian/observable encoding ($n = 1$). The main tradeoff across them is between width, depth, state-preparation cost, and the amount of linguistic structure preserved by the quantum circuit [20], [23], [29].

A. ANGLE ENCODING

Angle encoding is the dominant category in the corpus. In this scheme, a classical vector $\mathbf{x} = (x_1, \ldots, x_d)$ is mapped to rotation parameters,

$$|\phi(\mathbf{x})\rangle = \prod_{j=1}^{d'} R_{\alpha_j}(s_j(\mathbf{x}))|0\rangle^{\otimes n}, \tag{3}$$

where R_{α_j} is a rotation around the x, y, or z axis and $s_j(\cdot)$ denotes scaling, projection, or feature selection. The input

TABLE 1. Taxonomy of text encoding strategies in the final corpus: four major families and three emerging families.

Encoding family	Corpus count	Core mechanism	Typical model families
Major families			
Angle encoding	51	Text features are mapped to single-qubit rotation parameters, typically through R_x, R_y, or R_z gates.	VQC, hybrid neural, recurrent neural, quantum transformer
Amplitude encoding	28	A normalized feature vector is loaded into the amplitudes of an n-qubit state in a 2^n-dimensional Hilbert space.	VQC, quantum kernel, quantum transformer, hybrid neural
Syntactic/DisCoCat encoding	29	Sentence grammar determines a string diagram or circuit; word meanings are attached to grammatical types and composed by reductions.	Compositional QNLP
Structured feature maps	9	Classical text features parameterize structured circuits such as IQP, ZZFeatureMap, ZFeatureMap, PauliFeatureMap, or Pauli-Y variants.	Quantum kernel, VQC, hybrid neural, quantum transformer
Emerging families			
Data re-uploading	4	Input features are injected repeatedly across circuit layers, alternating data-dependent rotations with trainable transformations.	VQC, recurrent neural
Hybrid embedding	3	A classical encoder first produces a compact text representation that is then passed to a quantum layer.	Hybrid neural, VQC
Hamiltonian/observable encoding	1	Text-derived information is represented through Pauli-string Hamiltonians or observables rather than only through prepared quantum states.	VQC

is rarely raw text. It is usually a TF-IDF vector, a word or sentence embedding, a transformer representation, or a compact feature vector produced by a preceding classical model.

The prevalence of angle encoding follows from its implementation simplicity. Rotations fit naturally into parameterized quantum circuits and support the standard gradient-based training workflow of variational algorithms, and they avoid arbitrary-state preparation, which makes them convenient for small simulated or hardware-oriented experiments [21], [22]. What makes the family a single encoding category is this shared rotation mechanism; within it, the corpus studies differ along two encoding-level axes: the rotation construction itself and the classical source that supplies the angles.

The first axis is the rotation construction. The most common pattern assigns one feature to one single-axis rotation, typically an R_Y rotation as in the default angle-embedding routine [29]–[31], though R_X is also used [32]–[34] and some studies apply combined R_Y and R_Z rotations per qubit [35]. A denser variant stores more than one feature on each qubit through successive rotations about different axes, trading a shorter register for greater loading depth [5].

The second axis is the classical source of the angles, and it is where the studies vary most, because the rotation parameters are numbers rather than text and so the content of the encoding is fixed by the front-end that produces them. Some studies load sparse lexical statistics such as TF-IDF, bag-of-words, or n-gram vectors [36]–[38]; others load static word embeddings [31], [39], [40]; and a large, growing group compresses contextual transformer embeddings, for instance reducing a 768-dimensional BERT or DistilBERT vector to a handful of features before rotation [41]–[44]. A further group derives the angles from deeper classical feature extractors or from handcrafted linguistic features [45]–[47]. In each case a dimensionality-reduction step, whether PCA, a linear projection, or an autoencoder, sits between the text representation and the rotations and is effectively part of the encoding, since it determines what survives into the quantum circuit.

The limitation is inherent in this design. Encoding cost grows with the number of retained features, so a model must spend more rotations, more qubits, or more layers as it keeps more of the input. Angle encoding therefore pushes the hard part of the problem onto classical dimensionality reduction, which makes it practical for NISQ-scale prototypes but also means the quantum circuit often operates on a heavily compressed view of the linguistic input.

B. AMPLITUDE ENCODING

Amplitude encoding maps a normalized vector $\mathbf{x} \in \mathbb{R}^{2^n}$ or $\mathbb{C}^{2^n}$ to a quantum state,

$$|\phi(\mathbf{x})\rangle = \sum_{i=0}^{2^n-1} x_i |i\rangle, \quad \sum_i |x_i|^2 = 1. \tag{4}$$

For NLP, the appeal is immediate. A high-dimensional sentence embedding can, in principle, be represented with logarithmically many qubits: a 768-dimensional transformer vector, for example, can be padded and normalized into a ten-qubit state. This density is why amplitude encoding is most often paired with dense contextual representations, above all BERT, ELECTRA, and other transformer embeddings, and to a lesser extent with static word embeddings or TF-IDF and bag-of-words vectors [48]–[51].

Because the register holds exactly 2^n amplitudes, the encoding is defined as much by how the source vector is fitted to that dimension as by the loading step itself. When the embedding is already close to a power of two, it is simply normalized and zero-padded, as when a 768-dimensional vector is loaded into ten qubits [49], [52]. When it is not, or when the qubit budget is smaller, a classical reduction such as PCA, t-SNE, or UMAP is applied first, so that the reduction and the amplitude map together fix the geometry the circuit actually sees [53]–[55].

The reason to accept this encoding is representational. Storing a large vector in a small register makes overlap- and inner-product-based operations natural, because the normalized state is the embedding. SWAP-test similarity and quantum-kernel pipelines exploit this directly, letting the geometry of the encoded state stand in for the similarity structure of the embedding space [28], [51], [56]. A distinct

thread goes further and encodes complex-valued rather than real-valued amplitudes, representing tokens as complex state vectors and building complex-valued semantic operations on top of them, which uses a property of the amplitude space that has no direct classical-vector analogue [50], [57].

The signature difficulty is state preparation, and it is where the studies differ most in how honestly the cost is treated. Most rely on exact library routines that prepare an arbitrary amplitude vector, with the preparation depth absorbed into a simulator call [48], [49]. A smaller set confronts the cost directly through divide-and-conquer amplitude encoding, which trades circuit depth against ancilla width and is validated on `ibm_kyoto` for a word-sense disambiguation subset alongside the noiseless-simulation classification results [54]. Loading an arbitrary vector generally requires a circuit whose depth can offset the qubit savings, so experiments that hide this behind high-level routines remain useful for comparing representation strategies under a fixed qubit budget but should not be read as hardware-ready evidence unless the preparation cost is included.

C. SYNTACTIC AND DISCOCAT ENCODING

Syntactic encoding is the characteristic interface of compositional QNLP. Instead of first reducing a sentence to a flat feature vector, the model begins with a grammatical derivation. Words are assigned grammatical types, the sentence is reduced according to a pregroup or related grammar, and the resulting string diagram is translated into a tensor network or quantum circuit [24]. What makes the family a single encoding category is a shared two-stage construction: a diagram that fixes the circuit topology from grammar, followed by a rule that turns that diagram into a parameterized qubit circuit. The 29 studies in the corpus differ mainly along these two stages, and it is these differences, rather than the downstream task, that define the encoding-level variants below.

The first stage is how the diagram is obtained. The canonical route is a pregroup derivation, in which each word receives a pregroup type and grammatical reductions become the wires that connect word boxes; pregroup parsing is efficient and underlies most DisCoCat implementations [25]–[27]. In practice this parse is often produced by a broader-coverage CCG parser and then rewritten into DisCoCat form, as in studies that derive their diagrams with the Bobcat or DepCCG parsers before compilation [2], [58], [59]. A second variant replaces the parse with a fixed reader that imposes a uniform composition rule instead of syntax: a bag-of-words reader composes words through a single commutative spider, and a word-sequence reader composes them left to right into a tensor train, both of which appear as structure-free counterparts to the grammar-based diagrams [25], [60], [61]. These readers trade syntactic fidelity for shallower, more uniform circuits and are frequently used as structure-free reference points within DisCoCat studies. A third variant changes the scope of the diagram itself, lifting the sentence-level construction to the discourse level so that pronouns and cross-sentence reference are represented, as in DisCoCirc-style coreference models [62].

The second stage is how the diagram becomes a qubit circuit. Each atomic type is assigned a fixed number of qubits, so the noun and sentence types set the circuit width, grammatical reductions compile to Bell-style effects between wires, and function words such as relative or possessive pronouns are realized as Kronecker or GHZ spiders [27]. Word boxes are then filled by a parameterized ansatz, and the dominant choices are the IQP ansatz used by default in lambeq and, where circuit expressivity is compared, the Sim-14, Sim-15, and strongly entangling ansätze [9], [25]. Some studies design task-specific ansätze on top of the same grammatical skeleton rather than proposing a new diagram [1], [2]. A final variant intervenes between the two stages by rewriting the diagram before compilation: ZX-calculus is used to simplify the grammar-derived circuit and reduce its resource cost [63].

Unlike vector-loading schemes, the circuit here does more than host numerical features: its wiring reflects syntactic composition, so grammatical roles, reductions, and word interactions remain visible in the model structure. That interpretability explains why nearly all compositional papers in the corpus use this category, and its cost is flexibility. Syntactic encoding depends on parsers, grammar formalisms, and sentence structures that can be translated into manageable circuits, so many experiments use short, controlled, or grammar-friendly datasets. As sentences become longer, noisier, multilingual, or document-level, the parser and circuit-growth requirements become more restrictive.

D. STRUCTURED FEATURE-MAP ENCODING

Structured feature maps encode data through circuits with phase rotations, Pauli operators, and entangling gates whose parameters depend on products or interactions of input features, so that the encoded state defines an implicit feature space in which examples are compared by state similarity [28]. Within the corpus the map is drawn from a small set of families. The most common is the ZZ feature map, which introduces pairwise feature interactions through ZZ entangling rotations [64]–[66]. IQP embeddings, which apply diagonal phase gates in the Hadamard basis, are used similarly [67], [68]. A few studies construct higher-order maps from selected Pauli terms, for example combined X, Y, and ZZ gates or a hierarchy of Pauli-Y and second-order YY interactions, to encode richer correlations [68], [69]. One study instead reads the feature map out through local observables, using a projected quantum kernel to obtain a lower-variance similarity estimate [70].

Because both the qubit count and the number of encoded interactions grow with the input dimension, a structured feature map is almost always preceded by an aggressive classical reduction. In practice this is usually PCA to on the order of ten to a few dozen components, and occasionally a wavelet or related projection, applied to an already compressed text representation such as a TF-IDF vector or a reduced transformer embedding [67], [71], [72]. The reduction is thus part

of the encoding, since it fixes both the qubit budget and which feature interactions the map can express.

The benefit is that feature interactions are built into the map: rather than using the circuit only to host features, the map defines the geometry in which examples are compared, which makes the approach attractive for testing whether a quantum kernel yields a useful similarity space for text. That same design is also a strong modeling assumption, since performance depends on whether the chosen interactions are relevant to the linguistic task and on the preceding compression. A structured feature map therefore offers a principled kernel-oriented interface whose usefulness is set jointly by map design and the reduction that feeds it.

E. EMERGING ENCODING FAMILIES

The three remaining families are individually too small to rank among the major families; the taxonomy classifies them as emerging, each defined by a distinct mechanism. Data re-uploading addresses the small-register setting by injecting the input features multiple times across the depth of a circuit: rather than loading the data once and then applying only trainable gates, the model alternates data-dependent rotations with trainable transformations, so expressivity grows with depth rather than with qubit count. The four corpus studies differ mainly in what is re-uploaded and where. One implements a bricklayer of dense rotation gates in which repeated feature injection, rather than one-shot angle loading, is the defining encoding choice [73]. A recurrent variant re-uploads code-token embeddings across successive quantum embedding layers, so the same feature stream re-enters the circuit at every recurrent step [74]. A third injects lexical features repeatedly into a seven-qubit NISQ classifier, explicitly trading qubit width for circuit depth [75], and a fourth re-uploads sentence-transformer embeddings across the layers of an eighteen-qubit circuit, using qubit count and depth jointly to control capacity in a quantum LLM fine-tuning setting [76]. The shared advantage is that expressivity can be raised without more qubits; the shared drawback is that the encoding burden moves from width to depth.

Hybrid embedding is a system-level interface rather than a single gate-level rule. A classical model first produces a compact representation, and a quantum layer then consumes it through an angle, amplitude, or less explicitly specified feature-injection mechanism. In the corpus the three studies are distinguished mainly by how aggressively the classical stage compresses and how clearly the quantum loading is described. One compresses fused text and image embeddings into an eight-dimensional latent vector and matches it to an eight-qubit VQC for emotion recognition in dialogue [77]. Another compresses GloVe and fastText code-token embeddings into a minimal intermediate representation before a four-qubit vulnerability classifier, without specifying the gate-level loading rule [78], and a third fuses GloVe text features with VGG16 visual features through a quantum layer for misinformation detection, again with no stated state-preparation method [79]. This underspecification is characteristic of the family and feeds its main difficulty, attribution: when most semantic processing happens in the classical stage, the independent contribution of the quantum layer is hard to isolate.

Hamiltonian and observable encoding is even more specialized. The single corpus study maps text-derived vectors onto Pauli-string Hamiltonians and reads them out as quantum observables rather than preparing them as amplitude or rotation states, which avoids conventional state encoding entirely and is reported to keep qubit and gate complexity logarithmic [80]. This relocates the loading problem from state preparation to observable design and may sidestep some state-preparation costs, but with only one study in the corpus the evidence base is too thin to characterize the family beyond this single design point.

F. TAXONOMIC IMPLICATIONS

The taxonomy makes clear what an encoding choice commits a study to: which linguistic information reaches the quantum circuit, how much classical compression is required, and where the main resource bottleneck appears.

This distinction matters because many cross-paper comparisons implicitly compare different views of the input rather than only different learning architectures. A VQC using compressed word or sentence vectors, a DisCoCat circuit generated from a parser, and a kernel model built around a designed feature map make different commitments about what should be preserved from the original text. Their results are therefore shaped jointly by the classical representation, the compression method, the encoding circuit, the ansatz, the measurement rule, and the execution backend.

The taxonomy is therefore used in the remainder of the survey to organize the evidence, not to rank the encodings. The available studies do not yet show that any single family is generally superior across QNLP tasks; they show that the encoding stage frames the experiment itself, defining the form of the linguistic input, constraining the feasible circuit resources, and conditioning how performance and scalability results should be read.

V. COMPARATIVE ANALYSIS OF ENCODING FAMILIES

This section answers RQ2, using the taxonomy to assess how far the main encoding families have developed in practice. Rather than revisiting the definition of each family, the focus is on what the corpus reveals about their use: which encodings have become standard design choices, which remain exploratory, and how strongly the reported results are supported by datasets, baselines, hardware assumptions, and scalability analysis.

Throughout, the assigned encoding is only one stage of a pipeline that also fixes a text representation, a feature reduction, a trainable or fixed quantum model, and a readout rule, so an observed result is rarely attributable to the encoding alone.

The choice of encoding is not independent of the model family in which it appears. As shown in Figure 3, which

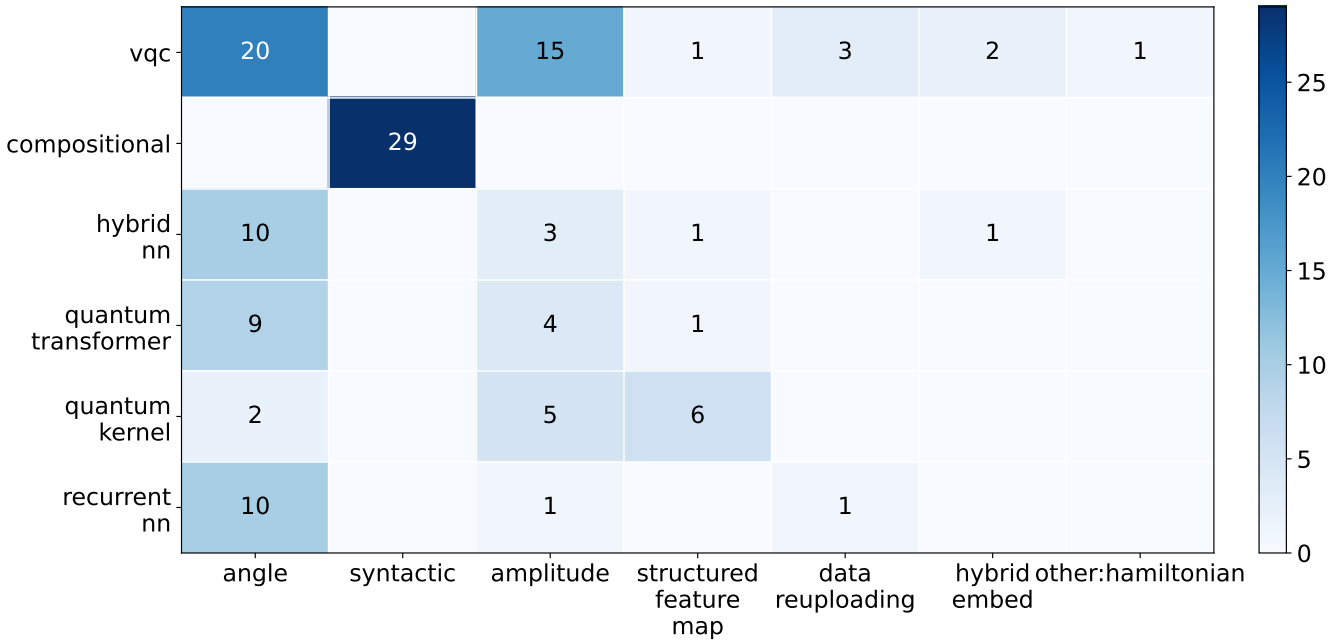


FIGURE 3. Encoding family by model family in the final corpus. Cell values report the number of studies assigned to each encoding–model-family pair.

crosses the two dimensions, the resulting distribution is far from uniform: most encodings concentrate in one or two model families rather than spreading evenly across them. The three dominant encodings account for 108 out of the 125 studies. One of the three, syntactic/DisCoCat encoding, coincides with the compositional model family by construction rather than by finding, since Section III-F identifies both categories by the same property, a circuit topology derived from a grammatical parse, so its 29 studies are the same 29 on either axis. The claim therefore has to rest on the remaining 96, and it holds there too, if less absolutely: every encoding family except angle places a majority of its studies in a single model family, with VQCs alone absorbing 42 out of the 96 and six of the nine structured-map studies sitting in quantum kernels. Angle encoding is the exception, the general-purpose default shared across the remaining gate-model architectures, with amplitude encoding co-residing alongside it in VQCs and kernels. Comparing "encoding families" across the corpus is therefore partly comparing the model families they are bound to. This motivates the approach of the four subsections that follow: each holds one established family fixed and asks how its use changes across the contexts in which it appears. The emerging families are treated last.

A. ANGLE ENCODING ACROSS MODEL FAMILIES

Angle encoding is the one interface that is not bound to a single architecture, so it is the natural place to begin: rather than asking which encoding each model family prefers, the analysis holds the encoding fixed and asks how the same data-dependent rotation is repurposed as the surrounding model changes. The 51 angle-encoding studies distribute across five model families, VQCs (20), hybrid neural models (10), recurrent models (10), quantum transformers (9), and quantum kernels (2), and their treatment of the rotation differs enough that "angle encoding" names a role rather than a fixed operation. As shown in Table 2, which summarizes the split, the remainder of this subsection reads it family by family. Qubit budgets are reported by 34 out of the 51 studies and, with one structural exception, stay within a narrow 2–8 band, so the differences below are differences in where the rotation sits, not in how wide a register it drives.

In VQCs, angle encoding appears in its canonical form: the rotation layer is the quantum front door, followed only by a trainable ansatz and a measurement, so the encoding carries the entire burden of moving text into the circuit. This is also the bucket with the widest spread of classical front-ends, which is why the compression the encoding demands is most visible here: TF-IDF or bag-of-words features, transformer embeddings cut by one or two orders of magnitude, from the 768 dimensions of BERT to ten, to four, or as far down as two values before an R_x layer, and occasionally handcrafted descriptors such as character entropy and diacritic counts [41], [42], [47], [81], [82]. Because the circuit is the model rather than a component of it, the VQC setting also hosts almost all of the controlled encoding comparisons in the corpus: a QCNN study finds angle encoding the best of angle, amplitude, and IQP maps [83], and a sentiment study finds an R_y angle map beating a Pauli feature map under matched circuits [29]. The rest of the bucket repeats this pattern unmodified while varying the task, the domain, and the readout: transfer learning on frozen BERT embeddings, AI-text detection, word-sense disambiguation, confidence cali-

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TABLE 2. Angle encoding split by model family in the final corpus.

Model family	Studies	Median (range) qubits	Typical classical front-end	Role of the angle rotation
VQC	20	4.5 (2–16)	TF-IDF, BERT/SBERT/DistilBERT, handcrafted features	Sole quantum front door before a trainable ansatz
Hybrid neural	10	4.5 (2–8)	Output of a classical transformer/RNN encoder	Submodule consuming an already-compact representation
Recurrent neural	10	4 (4–8)	Word or BERT embeddings, per token	Data input inside the recurrent cell, re-entered each step
Quantum transformer	9	4 (4–28)	Word embeddings, multimodal features	State preparation for quantum attention
Quantum kernel	2	4 (4)	TF-IDF or compressed feature vectors	Definition of a similarity-circuit geometry

bration, multimodal sentiment analysis, and privacy-oriented language modeling [84]–[91]. The numbers in Table 2 close the case for this bucket: a median of 4.5 qubits over a 2–16 range, the widest span of any row that loads its features once, because register width here is set by nothing but how far the classical front-end compresses. Attribution is cleaner than anywhere else in the family, at the price of a circuit that only ever sees that compressed view.

Hybrid neural models move the rotation downstream. The angle layer is no longer the front door but a submodule embedded in a larger classical network, so a transformer or recurrent encoder performs most of the representational work and the quantum circuit consumes a representation that has already been compressed classically. This is visible both in the front-ends, MARBERT, AraBERT, or XLM-R embeddings compressed into shallow circuits whose Pauli readouts are fused back into the classical model [43], a BERT-and-GRU multimodal pipeline projected into a two-qubit circuit, a transformer generator whose tokens are injected as rotations [92], and in the registers, which are the smallest in the angle subcorpus. The same downstream-submodule role recurs across the rest of the bucket, in fuzzy-neural sentiment layers, tensor-network fusion of text, audio, and visual streams, drop-in variational embedding layers, a quantum Mamba state-space model, and a GLUE-benchmarked quantum–classical BERT variant [93]–[98]. Hence the low end of Table 2, a 4.5-qubit median that never exceeds eight: the classical encoder has already absorbed the compression. The cost is the mirror image of the VQC case, as the independent contribution of the quantum rotation is the hardest of any family to isolate.

Recurrent models repurpose the rotation temporally rather than spatially. Here angle encoding is applied inside a quantum recurrent cell, a QLSTM or QRNN block, so the same rotation interface is re-entered at every step of the sequence rather than once at the input [30], [31], [35], [44]. This keeps the register deliberately narrow, a median of four qubits and never more than eight, because width is fixed by the cell while information enters through repeated application along the sequence, which brings this usage closer in spirit to data re-uploading than to a single loading step. The R_y-based AngleEmbedding is the recurring choice [30], and the bucket is the one most oriented toward token-level and sequential tasks, from part-of-speech tagging, including code-mixed tagging across Indian language pairs, to emotion recognition and sentiment monitoring over time [99]–[103]. Table 2 records the consequence, four qubits median and never above eight, the one row whose width is fixed by the cell rather than by the data.

Quantum transformers use the rotation to prepare the states that attention then compares, and this is the one place where the usual qubit discipline breaks. Features are mapped into per-token quantum states whose overlaps play the role of attention scores, so register width scales with sequence and embedding size rather than with a compressed feature count. QNet assigns one qubit per token–embedding-dimension pair and splits axes by role, with R_x carrying the embedding value and R_z a positional phase [32]. A near-term self-attention study reaches 28 qubits, the widest register in the entire angle subcorpus [104]. Other studies keep the per-token construction and vary only what the attention operates over, from graph message passing to mixed-density-matrix formulations and image–text fusion [105]–[108], and most avoid the blow-up by keeping the quantum attention block small [39], [109]. The 4–28 range in Table 2 is the signature of this bucket: it is the only row whose structural logic pushes angle encoding beyond the near-universal 4–8 qubit band.

Quantum kernels are represented by only two angle studies, and the contrast with the previous families is instructive. When angle encoding does appear in a kernel method, the rotation defines a similarity-circuit geometry rather than feeding a trainable ansatz [36], [110]; but kernel methods overwhelmingly prefer purpose-built structured feature maps (Section V-D), so angle encoding is a marginal, borrowed choice here rather than a native one. The exception is informative: angle encoding travels everywhere, but in the one family whose encoding is itself the model, a different interface wins.

Taken together, the split shows that holding the encoding fixed does not hold the pipeline fixed: the same rotation is a complete front door in VQCs, a downstream submodule in hybrid models, a per-step cell input in recurrent models, an attention-state preparation in quantum transformers, and a borrowed similarity map in kernels. These roles differ in exactly the dimensions that matter for evaluation, namely how much of the model the encoding accounts for, whether register width is set by compression or by architecture, and how cleanly the quantum contribution can be attributed. Only two things hold across all five rows of Table 2: the heavy classical compression that precedes the circuit, and the 4–4.5

TABLE 3. Amplitude encoding split by model family in the final corpus.

Model family	Studies	Median (range) qubits	Typical classical front-end	Role of the amplitude state
VQC	15	10 (9–16)	Normalized BERT/ELECTRA/DistilBERT/SBERT embeddings	Input state prepared for a trainable ansatz
Quantum kernel	5	10.5 (7–12)	Word/sentence embeddings, TF-IDF vectors	Overlap geometry read out by a SWAP test
Quantum transformer	4	7.5 (4–12)	Word or contextual token embeddings	State preparation for quantum attention
Hybrid neural	3	9 (6–12)	Multimodal or token embeddings	High-fidelity submodule after a classical encoder
Recurrent neural	1	—	Chinese word embeddings	Per-gate state inside a decomposed QLSTM

qubit median it produces in every row, quantum transformers included, whose 28-qubit outlier stretches the range without moving the center. This confirms the encoding–model-family correlation of Fig. 3 from the opposite direction, and it is why angle-encoded results are not directly comparable across architectures despite sharing a label.

B. AMPLITUDE ENCODING ACROSS MODEL FAMILIES

Amplitude encoding is the natural counterpoint to angle encoding, and reading its 28 studies through the same per-family lens makes the contrast concrete. It appears in the same host families, VQCs (15), quantum kernels (5), quantum transformers (4), hybrid neural models (3), and a single recurrent model, but with an almost inverted profile, both in which families it favors and in the resources it consumes. Where angle encoding compresses text to a handful of rotation parameters over a 2–8 qubit register, amplitude encoding loads a full normalized vector into the 2^n amplitudes of an n-qubit state, so its registers cluster near $\log_2$ of the embedding dimension rather than at a fixed small width. As shown in Table 3, which summarizes the split, the immediate signal is the qubit column: every family sits at a median near ten qubits, against the median of four seen for angle encoding, because the encoding is chosen precisely to avoid the aggressive projection that angle encoding requires.

VQCs are again the dominant host, but the role differs from the angle case. Amplitude encoding here prepares the input state that a trainable ansatz, typically EfficientSU2 or RealAmplitudes, then classifies [53]. The front-end is almost exclusively a dense contextual embedding normalized and padded to the nearest power of two, typically 768 SBERT, BERT, ELECTRA, or mBERT dimensions padded to 1024 amplitudes over ten qubits [48], [49], [52], with the $\log_2$ packing stated as the point of the method rather than an implementation detail [111]. Where a study compares the two interfaces directly, it frames amplitude encoding as the high-fidelity option and angle encoding as the cheaper fallback: one sentence-classification model reports its amplitude variant at 0.81 accuracy against 0.71 for an MLP-reduced angle variant of the same system [49]. The same host role recurs across the rest of the bucket, over tasks ranging from Italian-language sentiment transfer and spam detection to topic modeling and word-sense disambiguation in a low-resource ancient language [112]–[117]. Two numbers close this bucket. The first is in Table 3: ten qubits median over a tight 9–16 range, set by $\log_2$ of the embedding dimension rather than by any design choice. The second is temporal, as every VQC amplitude study appears from 2024 onward, tracking the point at which transformer embeddings became the default text representation and made a dense-vector encoding worth its qubit cost.

Quantum kernels are where amplitude encoding is most at home, in contrast to the marginal, borrowed status angle encoding held there. When a normalized embedding is amplitude-encoded, the inner product of two states equals the inner product of the underlying vectors, so a SWAP test or state-overlap estimate reads out cosine-like semantic similarity directly, with no training involved [50], [56], [118]. This is why the semantic-similarity task cluster of the corpus lives almost entirely in this cell, and why kernel studies apply the same construction to fastText, GloVe, or BERT embeddings and to TF-IDF vectors feeding a quantum nearest-neighbor classifier [51], [119]. The fit is principled rather than incidental, as the encoded state is the geometry the kernel measures, and some studies push the density further with complex-amplitude variants that map pairs of dimensions to the real and imaginary parts of each amplitude, halving the register again [50]. This is the one bucket where the qubit column of Table 3 records a deliberate investment rather than a cost: a 10.5-qubit median, the highest in the family, spent on keeping the embedding geometry intact because that geometry is the model.

Quantum transformers use amplitude encoding, as they use angle encoding, to prepare the states that attention compares, but over noticeably smaller and more variable registers than in the VQC case [120]–[124]. Amplitude encoding is rarer here than angle encoding (four studies against nine) because preparing an arbitrary state for every token is more expensive than injecting a rotation. The studies that do adopt it tend to encode a single pooled or sentence-level vector rather than a per-token sequence. Hybrid neural models place amplitude encoding in the same downstream-submodule role that angle encoding occupies there, but selected explicitly for fidelity: one knowledge-graph study states the trade-off within a single system, contrasting an amplitude model chosen for high-dimensional fidelity against an angle model chosen for gate efficiency [125], while others encode fused multimodal features or representations for natural-language-to-Prolog translation, the latter naming state-preparation cost as a limitation [126], [127]. Both buckets are small and sit below the family median in Table 3, at 7.5 and 9 qubits, which is the signature of encoding one pooled vector rather than a sequence.

The recurrent family is the sharpest point of the contrast.

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TABLE 4. Syntactic/DisCoCat encoding split by circuit ansatz in the final corpus.

Ansatz family	Studies	Median qubits (range)	Gate-level construction	Noun parameterization
IQP (LAMBEQ default)	19	7 (2–25)	Hadamard layers alternated with layers of parameterized controlled-R_z/CZ rotations, one block per grammatical reduction	Nouns are trainable rotations, same as other word types
Named alternative (SimpleSA, GeneralQC, DCA)	3	5 (4–10)	H–CNOT–R_z–H or H–CR_x/R_x–CNOT/R_x–CR_z blocks	Nouns fixed as identity gates or frozen embedding-derived states
Multi-ansatz benchmark (Sim14/15, StronglyEntangling, MPS/Spider/Tensor vs. IQP)	5	13 (7–20)	Same diagram compiled through several interchangeable circuit or tensor-network templates, compared head-to-head	Nouns trainable in every compared variant
ZX-calculus/underspecified	2	10 (single report)	Diagram or grammar tree compiled to a circuit via a generic, unlabeled routine; gate template not stated	Not reported

Amplitude encoding appears in exactly one recurrent study, a decomposed quantum LSTM over Chinese word embeddings [128], against the ten recurrent studies that use angle encoding, and the asymmetry is not accidental: re-preparing an arbitrary amplitude state at every step of a sequence is prohibitively expensive, whereas re-applying a rotation inside a cell is cheap. The encoding that is native to recurrent processing is therefore the opposite of the one that is native to overlap-based comparison. Read against the angle-encoding split, amplitude encoding partitions the same families by an opposite logic: angle encoding minimizes width and pays for it in classical projection, while amplitude encoding minimizes projection and pays for it in register width and, more importantly, in state-preparation depth. That last cost is the one most often abstracted away by simulator calls, so amplitude-encoded results are the ones whose reported qubit efficiency is most sensitive to whether the price of preparing the state has actually been counted.

C. SYNTACTIC/DISCOCAT ENCODING ACROSS ANSÄTZE

Syntactic/DisCoCat encoding cannot be split by model family the way the previous two families were, because all 29 of its studies belong to a single model family: compositional QNLP. This follows from the definition given earlier, since the grammatical reduction that wires the circuit is the model, so there is no separate architecture to vary. What does vary, and what plays the structural role model family played above, is the ansatz used to turn each grammatical box into gates, because that choice is the one genuine degree of freedom left once the parser has fixed the diagram. Two properties are shared across the whole subcorpus regardless of ansatz. First, the toolchain is a near-monoculture: 22 out of the 29 studies build their circuits with lambeq, and most of the rest use the related DisCoPy/pytket stack, in contrast to the diverse PennyLane, Qiskit, and TorchQuantum backends seen for the vector-loading families. Second, qubit count is not a function of the ansatz: it is fixed by the diagram, growing with sentence length and the number of grammatical wire types regardless of which gate template is dropped onto each box. As shown in Table 4, which groups the studies by ansatz, the qubit column shows exactly this: the reported range is nearly identical in the IQP bucket and in the small named-alternative bucket, two to twenty-five against four to ten, because register width tracks the sentence, not the gate design.

The IQP ansatz is the family's workhorse and accounts for two-thirds of the subcorpus. It originates in the generic ansatz families proposed alongside the first NISQ demonstrations and was then formalized as a depth-d word-circuit template, arity-one words as a single Euler rotation and higher-arity words as d layers of Hadamard-plus-neighboring-controlled-R_z gates, motivated explicitly by the conjectured classical hardness of IQP and later tuned with a cheaper single-rotation option for low-arity words [25]–[27]. This is the construction that lambeq packages as its default `IQPAnsatz` class [129]. The great majority of application papers in the corpus adopt it unmodified, changing only what is fed into the diagram or read out of the circuit: video and multimodal captioning, knowledge-graph relation modeling, sentiment classification, natural-language inference, optimizer benchmarking at fixed depth, diagram-reader ensembles, and pronoun resolution [3], [59], [62], [130]–[137]. The same block structure also travels across languages and modalities, to Chinese grammar, an English–Persian translation pipeline, and a musical-phrase model that treats snippets as words [4], [138], [139], and the one recurring modification is additive rather than structural, such as a negation gadget layered on an otherwise standard IQP circuit [58]. The 19 studies and the 2–25 qubit range of the IQP row in Table 4 therefore describe one template applied at many sentence lengths, not a design space being explored: once fixed by the toolkit, the ansatz is treated as infrastructure rather than as a variable.

A second, much smaller cluster treats the ansatz as the thing to redesign, and does so along one consistent axis: removing the noun from the pool of trainable gates. SimpleSA replaces the IQP block with an H–CNOT–R_z–H construction in which noun boxes are fixed identity gates rather than trainable unitaries, and reports higher accuracy than IQP with fewer parameters [1]. GeneralQC generalizes the idea to three concrete gate layouts: H-CR_x, R_x-CNOT, and R_x-CR_z. The two best-performing variants, R_x-CNOT and R_x-CR_z, are again the ones that leave nouns non-parameterized, reaching 96.91% accuracy against a 79.38% IQP baseline [2]. Parameter count does not track this pattern, however: both winning layouts use more trainable parameters than the IQP baseline (109 and 218 against 91), not fewer. The discrete circuit ansatz applies the same principle by a

different mechanism, replacing the trainable box of the noun with a fixed state derived from a classical word embedding through amplitude-style state preparation, and adding only a small trainable rotation keyed to the grammatical type, specifically to handle words the parser has never seen [140]. Across all three, the shared premise is that the syntactic role of a word, not its lexical identity, is what needs to be learned, and freezing the noun consistently improves or preserves accuracy on the tasks reported. It also cuts parameter count in SimpleSA specifically, but the best layouts of GeneralQC show this need not hold in general, since a redesigned ansatz can spend more parameters elsewhere while still gaining from non-parameterized nouns. What none of the three changes is the qubit column: at a median of five against the seven of IQP (Table 4), the gains here are bought in parameters, not in register width.

A third cluster asks the ansatz question directly, by fixing the diagram and swapping the gate template. The widest comparison pits lambeq's circuit ansätze, IQP, StronglyEntangling, Sim14, and Sim15, against its tensor-network ansätze, MPS, Spider, and plain Tensor, and finds Sim14 and Spider narrowly ahead of IQP among the circuit and tensor families respectively, with Sim15 essentially matching IQP and only StronglyEntangling clearly behind [9]. A second comparison of IQP against the Sim-et-al. "circuit 14/15" and StronglyEntangling ansätze finds the opposite depth dependence: StronglyEntangling is ahead at shallow depth, but IQP and Circuit14 converge faster and overtake it as circuit depth increases [141]. A related four-way comparison across six datasets gives IQP the best mean accuracy overall, while StrongEntangling wins on the hardest of them [61]. The same authors elsewhere present Sim14 and the Spider ansatz only as illustrative options, without running the comparison [142]. A narrower, non-lambeq comparison pits IQP against Sim14 directly on SQL circuits and finds Sim14 slightly more accurate but harder to train, concluding that the simplicity of IQP makes it the better default for that task [143]. Read together, these head-to-head studies complicate rather than confirm the workhorse status IQP holds by default elsewhere in the corpus. IQP is competitive rather than dominant: a hardware-efficient or tensor-network alternative beats it on accuracy or convergence speed in most, though not all, of the papers that run the comparison. It is also the ansatz that pulls ahead as circuit depth grows in the one study that varies depth explicitly. Its dominance in the rest of the subcorpus therefore reflects toolkit convenience at least as much as a demonstrated performance advantage.

Two studies leave the gate template unspecified, one routing the diagram through ZX-calculus rewriting before delegating compilation to a generic diagram-to-circuit call, with reduced qubit count as its only recoverable claim, the other building a custom pipeline whose parameterized boxes are never labeled [60], [63]. Neither permits the ansatz to be classified beyond "some parameterized construction," which is itself the finding for this row of Table 4: the least methodological transparency in the subcorpus falls on the one design choice this subsection is organized around.

Across the four buckets, syntactic/DisCoCat encoding is more homogeneous at the ansatz level than at the application level: a single IQP-derived template, inherited from the foundational papers of the family and packaged as lambeq's default, is adopted essentially unmodified by two-thirds of the subcorpus, whether the task is sentiment, translation, or music. The two genuine departures pull in different directions, one freezing nouns to cut parameters and the other benchmarking the template against hardware-efficient and tensor-network alternatives and repeatedly finding it beatable, and together they show the default gate template of the field to be a convenience inherited from tooling rather than a demonstrated optimum. What no row of Table 4 changes is qubit count, which stays governed by the grammar diagram itself [63], [143].

D. STRUCTURED FEATURE-MAP ENCODING ACROSS CONSTRUCTIONS

Structured feature-map encoding is the smallest of the four families addressed here, with nine studies, and it repeats the concentration seen for syntactic encoding: six of the nine are quantum-kernel methods, with only single instances in VQC, hybrid neural, and quantum-transformer models. Because model family again barely varies, the dimension that plays its role is the feature-map construction itself. That axis is natural for this family in any case. Unlike angle and amplitude encoding, which load feature values, a structured map builds feature interactions into the circuit through phase rotations and entangling gates; the choice of map is therefore a modeling assumption about which interactions matter. Two properties frame the split. Like angle encoding, the family depends on aggressive classical compression before the circuit: the high-dimensional text representation is reduced by PCA or a projection layer first. Unlike angle encoding, register width then tracks the number of retained features at roughly one qubit each. This is why the reported registers are comparatively wide, with a median of 10.5 qubits and a range of 5 to 20. As shown in Table 5, the studies are grouped by the map they use.

ZZ and IQP maps are the default construction, accounting for four of the nine studies. These are second-order maps that entangle pairs of features through data-dependent phase gates, and they are most at home in quantum kernels, where the encoded state defines the similarity geometry that the kernel measures: a ZZFeatureMap QSVM over a twenty-qubit register [64], an IQP feature map applied to reduced AraBERT embeddings [67], and IQP/ZZ maps feeding projected quantum kernels [70]. The same construction also appears once outside a kernel, as the primary quantum encoding layer of a hybrid model whose BERT/T5 embeddings are projected into a five-qubit ZZ map [65]. The 5–20 spread of this row in Table 5, the widest in the family, is exactly that borderline status made numeric: the same second-order map serves both as a kernel definition and as a swappable layer inside a trainable network.

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TABLE 5. Structured feature-map encoding split by map construction in the final corpus.

Map construction	Studies	Median qubits (range)	Interactions encoded	Host families and role
ZZ / IQP maps	4	10 (5–20)	Pairwise, second-order feature products via entangling phase gates	Mostly kernel geometry; one hybrid encoding layer
General Pauli maps	3	11 (6–14)	Higher-order and non-Z Pauli-string interactions	Kernel and VQC; gate combination chosen by search
Unnamed structured maps	2	not reported	Unspecified feature interactions	QSVM branch; transformer feature stage

General Pauli maps, in three studies, are the more expressive and more experimental construction. These use higher-order or non-Z Pauli strings and treat the exact gate content as a design choice to be tuned. A question-answering VQC selects a specific $X/Y/ZZ$ combination after a small search [69]. A quantum-enhanced SVM adopts a Pauli-Y/YYY feature map with a rotation factor over fourteen qubits produced by PCA [68], alongside an earlier Pauli-map sentiment kernel [71]. This is exactly where the central risk of the family becomes visible: because the map fixes which feature interactions are represented, performance depends on whether that hand-chosen structure happens to match the linguistic task, and the map is a strong assumption rather than a learned one. The eleven-qubit median of this row in Table 5, the highest of the three constructions, is the price of that assumption, since higher-order Pauli strings need one qubit per retained feature and offer no packing in return. The remaining two studies report only an unnamed "structured feature map", one as a QSVM branch compared against a VQC [66] and one as a feature stage inside an aviation-text transformer [72], a reporting gap that makes them hard to place against the named constructions.

Structured feature maps are the most-benchmarked yet least-dominant of the established families. Because the map is usually presented as one candidate encoding among several, the studies that use it tend to compare it directly with the vector-loading families, and those comparisons do not favor it: the group behind the Pauli-Y kernel above later reports that a simple R_y angle map outperforms a Pauli feature map under matched circuits [29]. Nine studies across three constructions, six of them in kernels (Table 5), leave the family a narrow but principled niche: the kernel column is its coherent core, where the map is the model rather than a replaceable layer, and its usefulness rests on a double dependency the other families do not share, the feature-map design and the compression stage that precedes it.

E. EMERGING FAMILIES

The residual group of eight studies comprises the three emerging families of the taxonomy, and unlike the four major families it cannot be split by model family for a different reason: it is already almost mono-cultural in architecture. Seven of the eight studies are variational or recurrent circuits, with a single hybrid-neural case and no kernel, compositional, or transformer instances at all. What varies, and what organizes this subsection, is the specialized mechanism itself, because each one is best understood as a targeted patch for a specific limitation of the mainstream encodings analyzed above rather than as a general-purpose alternative to them. As shown in Table 6, the studies are grouped by mechanism, naming the trade each one makes.

Data re-uploading, introduced in Section IV-E, is best read as the response to the width problem that constrains angle and amplitude encoding: re-injecting the same feature stream across circuit depth, rather than loading it once, lets a small register carry more information than a single loading step would allow. Its four studies span a six-to-eighteen-qubit range, wider than the fixed small band of angle encoding but nowhere near the $\log_2$-scale registers of amplitude encoding, because here the burden is shifted into depth rather than width.

Hybrid embedding is the response to the high-dimensionality problem, and by the same logic it carries the smallest registers in the emerging group, a median of six qubits (Section IV-E): a classical encoder absorbs the heavy compression before a small quantum layer ever sees the result. This is the same attribution problem already flagged for hybrid-neural angle encoding (Section V-A), taken to its limit: the independent contribution of the quantum tail is the hardest of any interface to isolate.

Hamiltonian and observable encoding, the single study covered in Section IV-E, is the response to the state-preparation problem that undermines the qubit savings of amplitude encoding (Section V-B): it reads text-derived information out through observables rather than preparing a state at all, sidestepping state preparation entirely while keeping a logarithmic, nine-qubit budget. With only one study, it remains an idea rather than an established interface.

The emerging families confirm the reading of the major families from the outside in: each is defined by the mainstream limitation it tries to escape, data re-uploading trading width for depth, hybrid embedding trading quantum responsibility for classical preprocessing, and Hamiltonian encoding trading state preparation for observables. Their shared confinement to variational and recurrent circuits shows that this experimentation is happening entirely within the trainable-circuit world, not in the kernel or compositional families whose encodings are structurally fixed. The evidence, however, is thinner and less comparable than for any established family: these studies differ in encoding and in where the quantum component sits and how much work precedes it. Three natural questions follow, one per family, and the current corpus answers none of them. Is added circuit depth worth the reduced register width? Does the quantum

TABLE 6. Emerging encoding families split by mechanism in the final corpus.

Mechanism	Studies	Median qubits (range)	Core mechanism	Host families and trade
Data re-uploading	4	7 (6–18)	Same features re-injected across circuit depth, alternating with trainable gates	VQC, recurrent; trades qubit width for depth
Hybrid embedding	3	6 (4–8)	A classical encoder emits a compact latent vector consumed by a quantum layer	VQC, hybrid; trades quantum role for classical preprocessing
Hamiltonian/observable	1	9	Data mapped to Pauli-string Hamiltonians or observables, with no state preparation	VQC; trades state encoding for observable encoding

layer contribute anything beyond the classical encoder that precedes it? Does reading information out through observables outperform preparing a state?

VI. TEMPORAL EVOLUTION OF ENCODING AND MODEL CHOICES

This section answers RQ3, reintroducing the dimension held constant so far: publication year. It establishes how large the field has grown and how its composition changed as it grew. The final corpus spans six publication years, from two studies in 2021 to a peak of 41 in 2025. The last calendar year is included at 27 studies but was not complete at the time of writing, since the search preceded the close of that year. Any apparent 2026 decline is therefore a search-cutoff artifact rather than a documented contraction, and it is treated as such throughout.

A. GROWTH AND THE CHANGING ENCODING MIX

Figure 4 decomposes the fivefold growth of the field by encoding family. Four movements account for it:

- Syntactic/DisCoCat recedes without shrinking. The only family present in 2021, it still holds half the corpus in 2022, then falls monotonically to 38% in 2023, 19% in 2024, and 11% in 2026, while its absolute count stays flat at three to eight studies a year. The decline is relative: the family did not contract, the field grew around it.
- Angle encoding absorbs most of the new volume. Its share rises from a quarter of the yearly corpus to roughly half after 2023.
- Amplitude encoding stabilizes with transformer embeddings. After a spike to a quarter of the small 2022 corpus and a drop below 5% in 2023, it settles at a quarter to a third of every year from 2024 on, the same year from which every VQC-hosted amplitude study runs on dense transformer embeddings (Section V-B).
- Everything else stays marginal. Structured feature maps remain in single digits apart from a one-off 14% in 2023, and the three emerging families together never exceed a tenth of the output of any year.

One explanation that could be sketched for this pattern is a difference in barrier to entry rather than in explanatory adequacy; no paper in the corpus frames its encoding choice in these terms, so it remains a hypothesis suggested by the aggregate counts. A DisCoCat study commits to a grammar formalism, a parser capable of producing pregroup or CCG derivations, and the narrow lambeq/DisCoPy toolchain, and pays that cost again on every new sentence, language, or domain (Section V-C). Angle and amplitude encoding ask only for a fixed-length numeric vector, so once pretrained embeddings became a routine input the marginal cost of a new study fell to whatever off-the-shelf VQC or kernel recipe was already circulating in the broader quantum-machine-learning literature. The families that stayed marginal are those that still demand bespoke engineering, a decomposed recurrent layer, a co-designed classical encoder, a custom observable construction, which scales with the number of researchers willing to do that work rather than with the size of the field.

B. FROM COMPOSITIONAL MONOCULTURE TO ARCHITECTURAL PLURALISM

Figure 5 tells the same story from the architecture side. Because compositional QNLP and syntactic encoding are the same 29 studies (Section V-C), the compositional decline, from the entire 2021 corpus to half of 2022 and 11% by 2026, is not independent evidence but the identical erosion seen on a second axis. What replaces that share is not one architecture but three:

- VQCs, the largest beneficiary. From an eighth of the 2022 corpus to about a third by 2025 and just under half in 2026. A VQC is not a QNLP-specific construction: Section V established it as the default host for both angle and amplitude encoding precisely because it asks nothing of the input beyond a numeric vector.
- Quantum transformers and kernels, steady but unspectacular. Both hold a seven-to-fifteen percent share from 2022 onward, with kernels dipping just below that band to 7.4% in the partial 2026 count. Each requires a deliberately quantum-native design decision, a state-overlap readout or an attention-like circuit structure, that cannot be copied from a generic VQC recipe.
- Hybrid neural models, the one outlier. Negligible through 2024, then 27% of the 2025 corpus, the single largest jump of any family in any year, before falling back to 4% in the partial 2026 count. This is also the architecture that reserves the smallest genuinely quantum role, and the one Section V-E flagged as hardest to attribute to its quantum component.

The 2026 column is partial throughout, so no fall recorded there can yet be read as a reversal. Taken with the encoding-level trend, what emerges is consistent on both axes: the

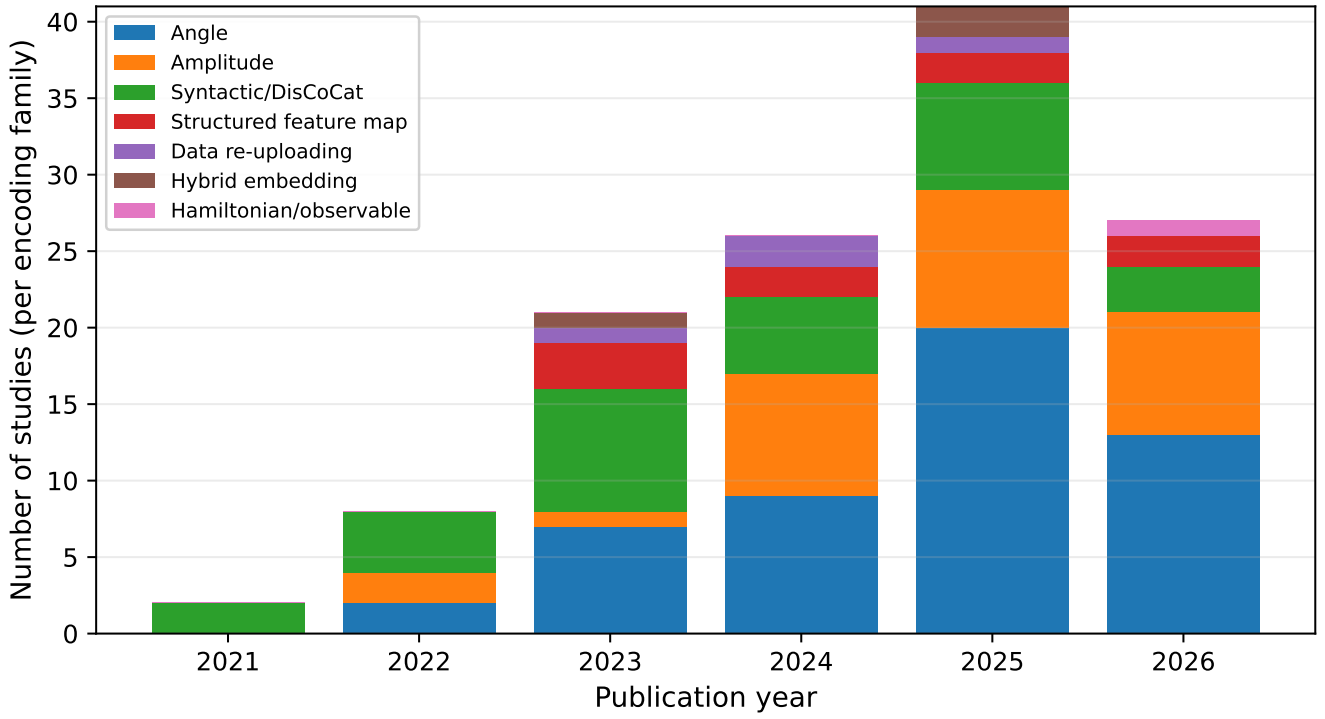


FIGURE 4. Encoding family by publication year in the final corpus (2026 is a partial year at the time of the search).

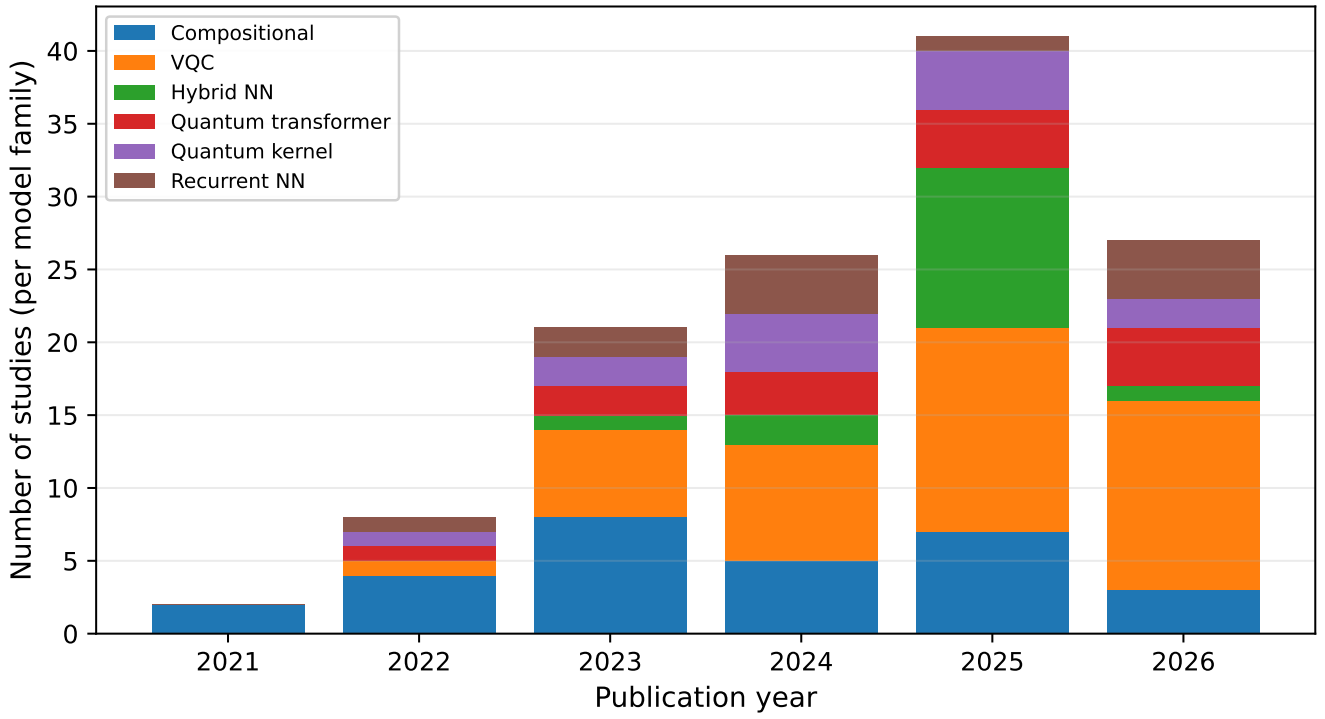


FIGURE 5. Model family by publication year in the final corpus (2026 is a partial year at the time of the search).

fastest-growing segments are those with the lowest entry cost for a researcher bringing a pretrained classical model

and a small amount of quantum machinery, not those built around a bespoke quantum-native design. On this reading, the growth of the field would have followed the path of least engineering resistance rather than the encodings and architectures judged most linguistically or computationally motivated in Section IV.

VII. TASK-LEVEL FIT BETWEEN ENCODING AND MODEL FAMILY

This section answers RQ4, examining whether the choice of encoding ever tracks the task itself, that is, whether any task family has a structure giving a principled reason to prefer one encoding over another, and whether the corpus shows the field acting on such a reason. The analysis is structural, framed in terms of what an encoding computes rather than in terms of reported results, for a reason set out at the end of the section: the corpus provides no basis on which the performance figures of two studies can be compared.

As Figure 6 shows, sentiment and other text classification together absorb the bulk of the corpus and every encoding family with any following at all. These two task families are nonetheless uninformative about task–encoding fit, since the encoding mix within each reproduces the corpus-wide mix: angle encoding takes 24 out of the 51 sentiment studies and 21 out of the 56 other-classification studies, amplitude 12 and 9, and syntactic 8 and 15, so the share of no family within either task departs from its corpus-wide value by more than about seven percentage points. No evidence in the figure indicates that classification selects for an encoding. The distribution is instead consistent with the corpus-wide habits documented in Sections V and VI, principally which toolchain and which pretrained embedding a group already had on hand, rather than with any property of sentiment or topic classification. With dozens of unrelated research groups applying broadly interchangeable recipes to broadly interchangeable binary and multi-class labeling problems, classification is the setting in which the corpus holds the most data and the least power to resolve the question.

Similarity and disambiguation is the one task family where a real, mechanistically grounded pattern emerges. Amplitude encoding accounts for six of the thirteen studies in the family, roughly double its share of the corpus overall, although on thirteen studies that enrichment is suggestive rather than established. The mechanism, not the count, is what carries the argument here, and Section V-B already established it: the inner product of an amplitude-encoded state literally is a cosine similarity, so a SWAP test or overlap measurement performs the core computation of the task rather than merely feeding a downstream classifier. The same mechanism explains why three of those six sit in the quantum-kernel model family [50], [56], [118]: a kernel method is little more than a trained readout wrapped around exactly that overlap. What the fit delivers, however, is feasibility rather than superiority, as those three studies themselves state. Their stated contribution is that the encoded state reproduces classical embedding similarity with minimal loss at logarithmic qubit cost, which is precisely what the mechanism predicts: an amplitude state loads the classical geometry it is given, so an overlap readout can reproduce a classical cosine similarity faithfully but has no channel through which to improve on it. The structural fit is real, and this is the one place in the corpus where the encoding provably does the work the task needs, but what it buys is efficiency rather than a better answer. The fit was moreover arrived at by mathematical necessity rather than by systematic search: no study in the corpus set out to compare encodings on similarity tasks and reached this conclusion, since it follows directly from the definition of amplitude encoding.

Relational tasks (question answering and, to a lesser extent, similarity itself) show a weaker echo of the same logic on the compositional side, but here the corpus is too thin to test anything. Syntactic/DisCoCat encoding takes two of the seven question-answering and retrieval studies and five of the thirteen similarity and disambiguation studies, which is at least consistent with the intuition that grammatical structure should matter most when a task is explicitly about how words or entities relate; on those denominators, however, the departure from the corpus-wide share of the family is not evidence of anything: one study either way moves the question-answering share by fourteen percentage points. The structural argument is in any case in tension with the practical constraint documented in Section V-C: the practicality of the family erodes precisely as a task departs from a short, curated sentence, and relational tasks are exactly where compositional structure is best motivated and exactly where the grammar-to-circuit pipeline is under the most strain. Six studies cannot settle which of the two effects dominates.

Sequence labeling, question answering, and generation or language modeling are each covered by only six to ten studies, too few to support a claim either way, but they fail differently. Sequence labeling is almost entirely angle-encoded, five of its six studies, so there is nothing to compare: one encoding occupies the cell rather than competing against alternatives inside it. Question answering and generation, by contrast, are spread thinly across several encodings, but each individual pairing is one or two papers from a single research thread rather than a pattern a second, independent group has reproduced. Neither failure mode supports a task-encoding claim, but for opposite reasons: one cell has no variation to explain, the others have too little replication to trust.

Three studies fall outside the six task families. They are reported here because the residual is small by construction rather than by neglect. Two are quantum topic models [38], [115]: both are unsupervised, evaluated by topic coherence and diversity rather than against labels, so they fit neither the supervised families nor generation; one uses angle encoding and the other amplitude, which on two studies settles nothing. The third is Quanthoven [139], which trains DisCoCat circuits to classify and then compose music; it is the one corpus member whose task is not linguistic at all, retained because its grammar-to-circuit pipeline is the same construction the compositional family uses on text.

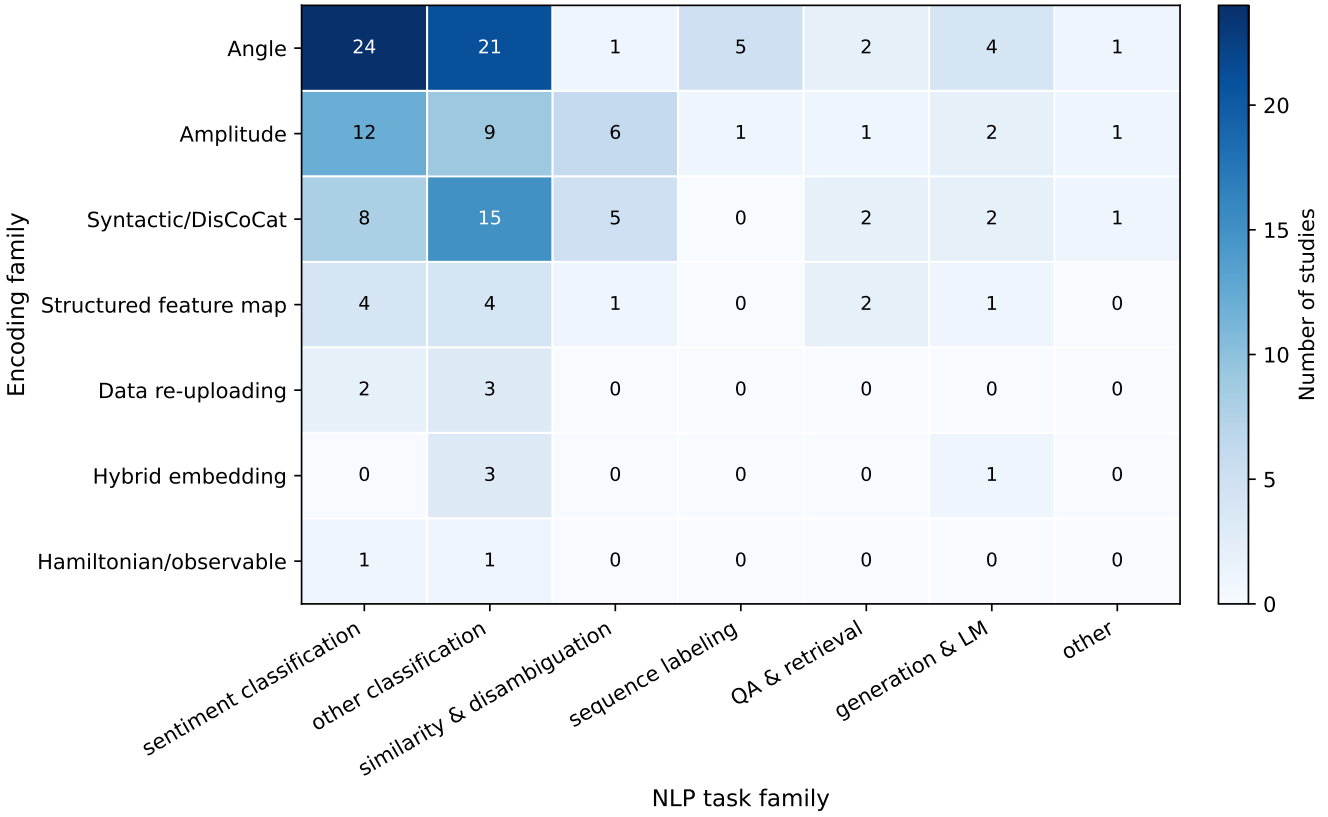


FIGURE 6. Encoding family by NLP task family in the final corpus. A study evaluating several task families is counted once in each.

This section's central finding is therefore a methodological one rather than a performance-based one: it is what licenses the structural reading used throughout. The corpus cannot be pooled on results. Each study selects its own dataset, its own classical comparison, and its own evaluation protocol: 141 distinct datasets are named across the 125 studies, and 122 of them are used by exactly one paper (Section VIII-D); the classical systems compared against are nearly unique to each paper as well; and 15 studies report no classical baseline at all. A reported accuracy figure in this corpus is thus a statement about one pipeline, on one dataset, against one baseline; the difference between two such figures, drawn from two different studies, is not attributable to the encoding that distinguishes them. This is why the one task–encoding fit that the corpus does support has to be argued from mechanism rather than measured from results, and why even that fit resolves into efficiency rather than superiority. Determining whether an encoding does better on a task, rather than merely fitting it, requires the shared benchmarks called for in Section IX-D.

VIII. EXPERIMENTAL RIGOR: SIMULATION, HARDWARE, AND REPRODUCIBILITY

This section answers RQ5 by assessing how rigorously the corpus tests the encoding and model choices catalogued so far. Every study was coded for its execution regime (noiseless simulation, noisy simulation, real hardware, or a mixed protocol), the simulation framework and backend used, code availability, and whether scalability was discussed. Together these fields answer a question the preceding sections cannot: not which encoding is used, but how much evidence backs the claim that it works.

A. EXECUTION REGIMES ACROSS THE CORPUS

Noiseless statevector simulation dominates: 96 out of 125 studies (76.8%) report exact simulation only. A further 16 (12.8%) simulate under an explicit noise model, and 11 (8.8%) report a mixed protocol pairing simulation with a hardware validation run. Only two studies (1.6%) run exclusively on real hardware with no simulated baseline: a quantum-kernel KNN classifier [51] and a quantum self-attention model on a 72-qubit superconducting processor [104]. Combining the mixed and hardware-only categories, 13 studies (10.4%) are hardware-touched; widening this to any engagement with noise (noisy simulation, mixed protocol, or hardware-only) raises the count to 29 studies (23.2%), termed noise-or-hardware engagement. The remaining three-quarters of the corpus never leaves a noiseless simulator. Hardware-touched status required the study to name the device it ran on: three studies that report a quantum backend

TABLE 7. Execution regime by encoding family in the final corpus.

Encoding family	Noiseless	Noisy sim	Mixed	HW-only
Angle	42	6	2	1
Amplitude	24	2	1	1
Syntactic/DisCoCat	19	6	4	0
Structured feature map	5	1	3	0
Data re-uploading	2	1	1	0
Hybrid embedding	3	0	0	0
Hamiltonian/observable	1	0	0	0

but resolve, on inspection, to a fake-hardware noise model or a vendor simulator are coded here as noisy or noiseless simulation rather than as hardware runs.

Table 7 crosses execution regime with the taxonomy of Section IV. As a whole the table shows no clean split between the two dimensions, and four of the seven families contribute fewer than ten studies each, so the per-family percentages below are description rather than demonstrated difference, except for the one contrast at the end of this paragraph that rests on denominators large enough to carry weight. Descriptively, engagement with noise and hardware concentrates in the families Section V already identified as small and toolchain-bound. Syntactic/DisCoCat encoding is hardware-touched somewhat more often than the corpus average (13.8% against 10.4%) and engages noise or hardware at more than twice the rate of angle and amplitude encoding (34.5% against 16.5%), because its lambeq/DisCoPy/pytket stack (Section V-C) ships with direct IBM Quantum integration and its shallow, few-qubit circuits are cheap to queue on real devices [25], [27], [61], [139]. Structured feature maps and data re-uploading show the highest hardware- and noise-touched shares of all, but on denominators of nine and four studies, so these describe a handful of papers rather than a trend. The contrast that does hold: angle and amplitude encoding, 79 out of the 125 studies, engage noise or hardware at less than half the rate of the rest of the corpus (13 out of 79, 16.5%, against 16 out of 46, 34.8%). Restricted to real-device runs the asymmetry points the same way, on smaller numbers (5 out of 79, 6.3%, against 8 out of 46, 17.4%). Since these two encodings carry the growth of the field (Section VI-A), most of their reported accuracy numbers (66 out of 79) are simulator artifacts, numbers silent on whether the same circuit would survive decoherence, gate error, or readout noise on real hardware.

Hybrid embedding and Hamiltonian/observable encoding report no hardware or noisy-simulation validation whatsoever: every study in both families is exact-simulation only. For Hamiltonian encoding this is one study, so no pattern can be claimed, but it means the central promise of the family, that observable-based readout sidesteps the state-preparation cost of amplitude encoding (Section V-E), has not been tested under any condition that would actually stress it.

B. EXECUTION BY MODEL FAMILY AND THE RECURRENT-NETWORK BLIND SPOT

The same asymmetry appears when execution is crossed with model family instead of encoding. Quantum kernels are the most hardware-touched family (3 out of 13 run on a named device, 23.1%), consistent with their concentration in amplitude and structured-feature-map encoding (Section V-B); on the wider measure compositional models lead instead, 34.5% of them engaging noise or hardware against the kernels' 30.8%, the same figure as syntactic encoding, since the two categories are the same 29 studies. VQCs, hybrid-neural models, and quantum transformers cluster around 19–27%. Recurrent quantum models are the one family with zero: all 12 recurrent-network studies (QLSTM, QRNN, and related sequential architectures) report noiseless simulation only. This matters because a recurrent quantum cell re-applies its encoding and ansatz once per token (Section V-A), so circuit depth, and exposure to decoherence and gate error, scales with sequence length in a way a single-shot VQC does not: the family whose central architectural claim rests most on reusing a small register cheaply is the one for which noise robustness has been tested least.

C. HARDWARE VALIDATION PEAKED IN 2024 AND THEN COLLAPSED

Section VI-A showed the corpus grew roughly twenty-fold between 2021 and 2025, but the hardware-touched share does not grow with it. It climbs slowly to a peak and then collapses: 0% in 2021, 12.5% in 2022, 14.3% in 2023 and 26.9% in 2024, then 2.4% in 2025 and 3.7% in 2026 (partial year). Noisy-simulation share does not follow the collapse, holding between 12% and 15% in every year from 2022 onward except 2024, so the late-period loss is specific to real-device runs. Pooling the window puts the size of the drop plainly: 11 out of 57 studies published 2021–2024 are hardware-touched (19.3%), against 2 out of 68 in 2025–2026 (2.9%). The matching contrast for noise-or-hardware engagement generally is far milder, 17 out of 57 (29.8%) against 12 out of 68 (17.6%), consistent with real-device execution specifically falling away rather than noise modeling. Setting aside the two-paper 2021 cohort, the two most recent years thus show the lowest combined rate of hardware or noise engagement of the entire window, 17.1% and 18.5% against 25–35% across 2022–2024, at the exact moment the output of the field peaked, the experimental-rigor counterpart to the low-barrier mechanism Section VI-A proposed for the encoding mix: growth ran through the protocol that asks the least of a newcomer, a local noiseless simulator call rather than a queued hardware job or a calibrated noise model. Two studies reach hardware anyway: a neural-quantum-embedding pipeline validated on an IBM backend [65], and a quantum-secure federated-learning system validated on a 156-qubit backend, though motivated by communication security rather than an NLP-performance claim [144].

D. FRAMEWORK FRAGMENTATION AND THE REPRODUCIBILITY GAP

Even where a study stays in simulation, the tooling underneath it is far from standardized. 37 out of 125 studies (29.6%) do not name the simulation framework at all, the

single largest category. Among those that do, PennyLane is most common (21 sole and 22 more in combination, 43 studies in all), Qiskit is second (9 sole, 25 mentions in all), and lambeq anchors the syntactic/DisCoCat subcorpus (Section V-C); the rest (TensorCircuit [111], Amazon Braket Hybrid Jobs, Qulacs, TensorFlow Quantum, Flax) are each unique to a single paper. Concentration in the corpus is therefore at the framework layer alone: outside the lambeq cluster and a PennyLane/Qiskit core, the stacks built on top of those frameworks are close to unique per paper. This compounds a second gap: only 25 out of 125 studies (20.0%) report publicly available code. Between the two gaps, 34 studies (27.2%) report neither code nor a framework and only 22 (17.6%) report both, so a substantial fraction of the corpus cannot be independently re-run even in simulation, let alone re-validated on hardware.

Scalability discussion fares better on paper: 91 studies (72.8%) discuss it against 34 that do not. But like the evaluation protocols of Section VII, this is self-reported against whatever standard the authors of each paper chose, ranging from a formal resource analysis to a single sentence deferring qubit count to future work. Combined with the noiseless-simulation dominance of Section VIII-A, it remains unclear whether any encoding in Section IV would still work outside a simulator at a larger register size, generalizing the state-preparation caution Section IV-B raised specifically for amplitude encoding.

IX. TRENDS AND OPEN CHALLENGES

This section reads Sections IV–VIII together, states plainly what the encoding bottleneck named in the abstract is, and converts the gaps into an agenda for future work.

A. WHERE THE FIELD IS HEADING

Three trends established separately above describe a single movement. The field is consolidating on the encodings that ask least of a newcomer, angle and amplitude, at the expense of the grammar-dependent syntactic/DisCoCat tradition (Sections VI-A–VI-B). That consolidation is landing on simulators rather than on hardware: 2025 and 2026, the years the corpus grew fastest, carry the lowest combined rate of hardware or noisy-simulation engagement of the entire six-year window (Section VIII-C), and the shortfall concentrates in the same two families that carry the growth (Section VIII-A). And the one mechanistically grounded task–encoding fit in the corpus, amplitude encoding on similarity and disambiguation, is small, drawn disproportionately toward quantum kernels, and arrived at by mathematical necessity rather than by systematic search (Section VII).

These are not three independent observations: the growth curve of the field and its evidentiary curve are decoupling. Publication volume is rising fastest in exactly the segment least equipped to demonstrate anything about noise, hardware behavior, or a genuine quantum contribution, while the segment with the clearest structural argument for one remains a slow-growing niche. Underneath sits the incommensurability documented in Section VII, which leaves a reported figure carrying little evidential weight beyond the single pipeline that produced it. None of these gaps belongs to one encoding or one model family; they are properties of how the field as a whole reports its experiments, which is why closing them is a precondition for the task–encoding questions to be answerable at all.

B. THE ENCODING BOTTLENECK

Every family reviewed in Sections IV and V but one is, underneath its individual mechanism, a different response to the same arithmetic. Let d denote the dimensionality of the text representation a study starts from, in this corpus from a handful of TF-IDF or handcrafted features up to the 768 or more of a transformer embedding, and let q denote the qubit budget it actually uses, empirically two to sixteen across the angle and amplitude subcorpora (Tables 2 and 3). Angle encoding spends one rotation on each retained feature, so the register must be about as wide as the feature count $d' \leq d$ that survives a classical reduction applied before the circuit is built: the mismatch is resolved by discarding information, and the denser variants of Section IV-A narrow the register only by paying in loading depth. Amplitude encoding instead packs the full vector into $q = \lceil \log_2 d \rceil$ qubits, but preparing an arbitrary 2^q-dimensional amplitude vector generically requires a gate count scaling with d rather than with q [145], [146] (Section IV-B); here the mismatch is paid in circuit depth, a cost Section V-B showed is routinely absorbed into a simulator call and left out of the reported qubit budget. Structured feature maps and data re-uploading sit between the two, trading some width for some depth without eliminating either cost, and hybrid embedding relocates the mismatch upstream into a classical encoder rather than resolving it. Syntactic/DisCoCat encoding alone sidesteps the arithmetic instead of trading inside it: q is set by the grammatical structure of the sentence, not by the dimensionality of an embedding (Section IV-C), which is why Section IV treats it as categorically different and why it is the one encoding whose qubit count Section V-C found untouched by ansatz choice.

This is the encoding bottleneck named in the abstract: a persistent gap between d, set by what classical NLP representations require to be informative, and q, set by what near-term hardware can support. Every non-syntactic family is best understood as a choice of which cost absorbs the gap, lost information, added depth, or a relocated classical burden, rather than as a choice that closes it, and no study in the corpus reports an encoding that escapes all three. The bottleneck is structural, not a matter of engineering effort within any single family.

C. OPEN CHALLENGES

The gaps documented throughout the review reduce to five open challenges, each derived from the corpus itself rather than imported from general quantum-computing commentary.

C1) The encoding bottleneck and the information it costs. Outside the syntactic family the reduction stage discards most of the representation before the circuit ever sees it, 768-dimensional embeddings cut to ten [41], four [42], or two [82] features, while state-preparation costs are routinely left out of the reported qubit budget. The mitigations reported in the corpus, divide-and-conquer preparation [54], observable encoding [80], and depth-for-width trades [75], change which cost is paid without closing the gap, and no surveyed study quantifies what the reduction removes.

C2) Incommensurable evaluation. No two studies can be compared on results, for the reasons set out in Section VII. Even the one structurally motivated pairing, amplitude encoding on similarity tasks, is presented by its own studies as reproduction of classical similarity rather than an improvement on it [50], [56]. Open because no shared benchmark with fixed classical baselines exists against which results from different studies could be made commensurable.

C3) The simulation–hardware gap. 96 out of 125 studies never leave a noiseless simulator; real-device validation peaked in 2024 and then collapsed to at most 3.7% in 2025–2026 (Section VIII-C), the only hardware-validated studies of those two years being [65], [144], the latter reaching hardware for reasons orthogonal to an NLP-performance claim. Open because noiseless simulation is silent on decoherence, gate error, and readout noise, precisely the properties that decide near-term viability.

C4) Absent scaling evidence. 102 out of 125 studies evaluate a single register width; only 23 test more than one, and systematic sweeps such as the 1–16-qubit comparisons of [54], [76] are rare. Open because every resource argument in Section IX-B is asymptotic while the evidence is single-point.

C5) Reproducibility and benchmark triviality. 100 studies (80.0%) release no code and 37 (29.6%) do not name their simulation framework (Section VIII-D); of the 141 datasets named across the corpus, only 7 recur in four or more papers. Those that do recur are either generic sentiment benchmarks carrying no QNLP-specific signal (IMDb, 12 papers; Amazon reviews, 8; Yelp, 7) or the ∼100-sentence synthetic MC/RP toy corpora built into lambeq (12 and 9 papers), on which near-perfect accuracies are reported [25], [60], [136]. Open because at this reporting standard neither independent re-execution nor a shared benchmark is possible.

D. AN AGENDA FOR FUTURE WORK

Four actions follow directly from C1–C5.

Noise-in-the-loop evaluation as a default. Report accuracy under an explicit, calibrated noise model as a minimum bar for publication, rather than treating noise as a qualitative caveat. The case follows from C3: angle and amplitude encoding are simultaneously the fastest-growing families and the least hardware- or noise-validated, and recurrent quantum architectures, which re-apply their encoding once per token, have no noisy-simulation or hardware study anywhere in the corpus (Section VIII-B).

Controlled, cross-family encoding comparisons. Hold model family, dataset, and circuit depth fixed while swapping only the encoding or ansatz, making the controlled comparison the template rather than the exception. Section V showed encoding and model family to be so tightly coupled that comparing "encoding families" is often really comparing the architectures they are bound to; the few studies that already run such comparisons are disproportionately informative relative to their count.

Shared benchmarks with fixed classical baselines. Establish a small set of shared, encoding-agnostic NLP benchmarks with fixed classical baselines, of the kind proposed piecemeal but never standardized in the surveyed studies. This addresses C2 directly, and would let a future meta-analysis attribute a performance gap to the encoding rather than to the baseline an individual group selected.

From structural fit to a measurable gain. Search deliberately for task families where the mathematical structure of an encoding performs the core computation of the task, and prefer fits where that computation is classically expensive: whether the compositional structure of syntactic encoding can be matched to relational tasks such as question answering, for instance, where Section VII found a theoretical fit the corpus is too thin to evaluate. The one documented fit yields efficiency rather than a better result, since the encoded state reproduces exactly the classical geometry it loads. A targeted study, rather than another general-purpose classifier, could establish whether a structural fit exists anywhere that also produces a gain measurable against a fixed classical baseline.

X. CONCLUSION

This article treated text encoding as the primary variable of QNLP research. Under a protocol fixed before the search, 376 database records and 1222 snowball candidates were screened down to 125 studies published between 2021 and 2026, organized under a seven-family taxonomy defined by the point at which text enters the quantum computation.

Four findings recur. Encoding and model family are tightly coupled, so comparing encoding families is frequently a comparison of the architectures they are bound to. The rapid growth of the field since 2023 has been a consolidation onto the encodings that ask the least of a newcomer, at the expense of the grammar-dependent compositional tradition. That growth has not been matched in experimental rigor, as three-quarters of the corpus never leaves a noiseless simulator and real-device validation peaked in 2024 and then collapsed. Results cannot be pooled, since dataset, baseline, and protocol are chosen study by study; the single task–encoding fit the corpus supports is argued from mechanism, and yields efficiency rather than a better answer.

Underlying all four is the encoding bottleneck: the persistent gap between the dimensionality of a useful text representation and the qubit budget near-term hardware can support. Every surveyed family except syntactic encoding pays for that gap in discarded information, added circuit depth, or a relocated classical burden, and none eliminates it. Three caveats condition these results. The search covered Scopus and IEEE Xplore and, although two rounds of backward and forward snowball sampling were used to recover work those databases miss, seeding from the surveys of the field leaves work with no citation link to any survey outside its reach. The 2026 counts are partial, since the search was conducted before the year closed, so the final point of every year-over-year trend should be read with that in mind. A few extracted fields, scalability discussion among them, are self-reported by the studies themselves rather than independently re-derived.

Text encoding is therefore not a neutral preprocessing step, and it warrants the scrutiny given to the ansatz, the optimizer, or the classical baseline.

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